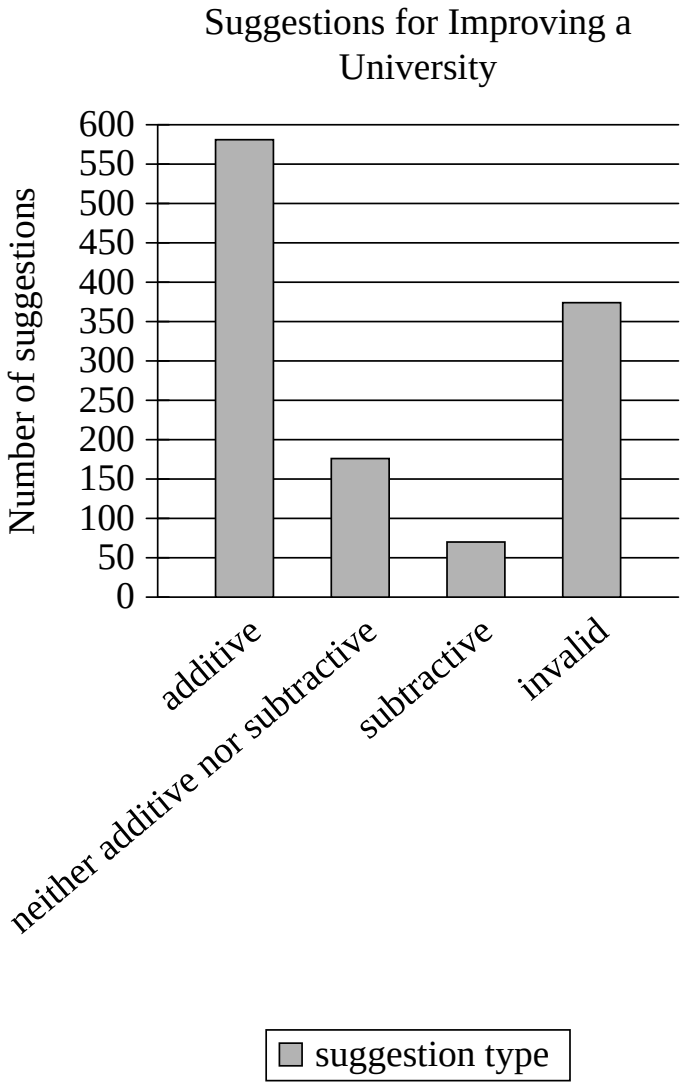
Command of Evidence — Part 2

Information and Ideas · Reading and Writing · 93 questions



- The data for the 4 categories are as follows:
  - additive: 581 suggestions
  - neither additive nor subtractive: 176 suggestions
  - subtractive: 70 suggestions
  - invalid: 374 suggestions

Gabrielle Adams and colleagues reviewed suggestions for improving a university that had been submitted to the university's president. They coded each suggestion as additive (the idea suggested adding something new to the university), subtractive (the idea suggested removing something from the university), neither additive nor subtractive, or invalid (the idea was not comprehensible). The data illustrated people's tendency to overlook the possibility of removing things to achieve improvements: _____blank

Which choice most effectively uses data in the graph to complete the statement?

A. around 175 suggestions were coded as neither additive nor subtractive, whereas around 575 suggestions were coded as additive.

B. more than 350 suggestions were coded as invalid, whereas fewer than 100 suggestions were coded as subtractive.

C. fewer than 100 suggestions were coded as subtractive, whereas more than 550 suggestions were coded as additive.

D. around 575 suggestions were coded as additive, whereas around 175 suggestions were coded as subtractive.

2.2 62962b25-dd31-433c-93d0-1032c533c58d · Medium

“Mr. Cornelius Johnson, Office-Seeker” is a 1900 short story by Paul Laurence Dunbar. In the story, the narrator describes Mr. Cornelius Johnson’s appearance as conveying his exaggerated sense of his importance: _____blank

Which quotation from “Mr. Cornelius Johnson, Office-Seeker” most effectively illustrates the claim?

A. “He carried himself always as if he were passing under his own triumphal arch.”

B. “The grey Prince Albert was scrupulously buttoned about his form, and a shiny top hat replaced the felt of the afternoon.”

C. “It was a beautiful day in balmy May and the sun shone pleasantly on Mr. Cornelius Johnson’s very spruce Prince Albert suit of grey as he alighted from the train in Washington.”

D. “Mr. Cornelius Johnson always spoke in a large and important tone.”

2.3 b4fd3c61-604e-4f87-b027-41ddf01d205a · Medium

Fish whose DNA has been modified to include genetic material from other species are known as transgenic. Some transgenic fish have genes from jellyfish that result in fluorescence (that is, they glow in the dark). Although these fish were initially engineered for research purposes in the 1990s, they were sold as pets in the 2000s and can now be found in the wild in creeks in Brazil. A student in a biology seminar who is writing a paper on these fish asserts that their escape from Brazilian fish farms into the wild may have significant negative long-term ecological effects.

Which quotation from a researcher would best support the student's assertion?

A. "In one site in the wild where transgenic fish were observed, females outnumbered males, while in another the numbers of females and males were equivalent."

B. "Though some presence of transgenic fish in the wild has been recorded, there are insufficient studies of the impact of those fish on the ecosystems into which they are introduced."

C. "The ecosystems into which transgenic fish are known to have been introduced may represent a subset of the ecosystems into which the fish have actually been introduced."

D. "Through interbreeding, transgenic fish might introduce the trait of fluorescence into wild fish populations, making those populations more vulnerable to predators."

2.4 293c8b41-0e97-4441-91c2-93422b4dbb23 · Medium

Water flowing around an obstruction creates vortices (patterns of swirls) of varying size; by detecting the vortices, fish can determine the size and position of the obstruction. Testing by Yuzo R. Yanagisuru, Otar Akanyeti, and James C. Liao using models of three head shapes—narrow (low ratio of width to length), intermediate, and wide (high ratio of width to length)—showed that for medium-sized vortices, fish with wide heads would be least able to distinguish between vortices and general turbulence in the water. A second research team has therefore hypothesized that in low-visibility conditions, wider-headed fish will be less likely than narrower-headed fish to detect obstructions.

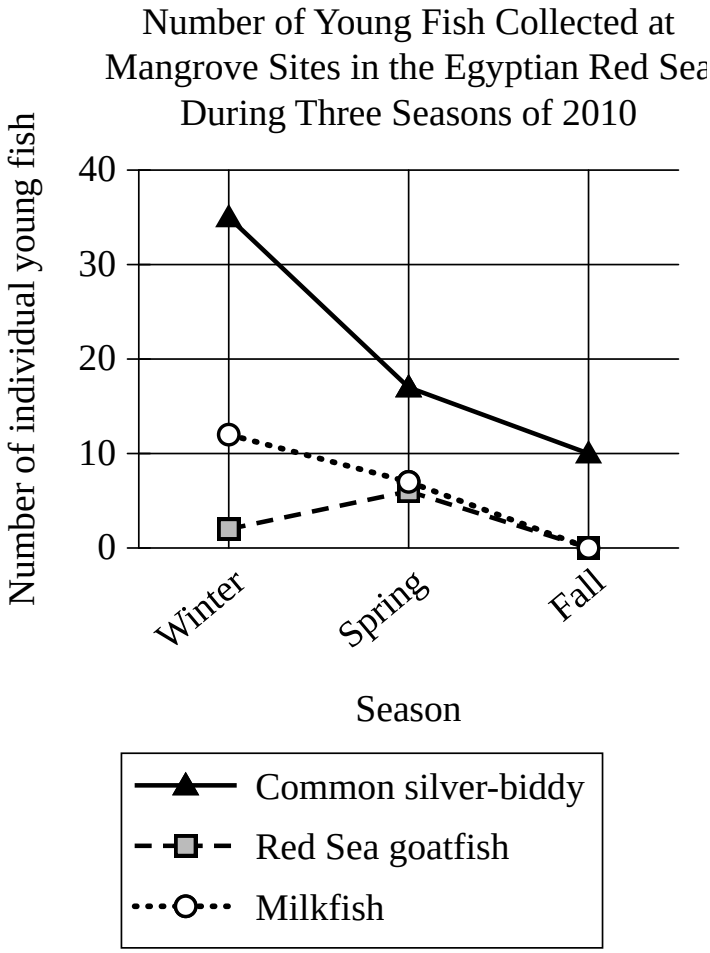
Which finding, if true, would most directly support the second research team's hypothesis?

A. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the bristlemouth (*Chaetostoma yurubiense*), which has a relatively wide head, bumped into more than half of the obstructions.

B. A study using obstructions that created medium-sized vortices in low-visibility conditions found that some specimens of dusky smooth-hound (*Mustelus canis*), which has a relatively narrow head, bumped into the obstructions more often than other specimens of the same fish did.

C. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the wider-headed bristlemouth (*Chaetostoma yurubiense*) bumped into obstructions more often than the narrower-headed dusky smooth-hound (*Mustelus canis*) did.

D. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the narrower-headed dusky smooth-hound (*Mustelus canis*) bumped into the obstructions just as often as the wider-headed bristlemouth (*Chaetostoma yurubiense*) did.



- The following 3 lines are shown:
  - Common silver-biddy
  - Red Sea goatfish
  - Milkfish
- The Common silver-biddy line:
  - Begins at Winter, 35 fish
  - Falls sharply to Spring, 17 fish
  - Falls gradually to Fall, 10 fish
- The Red Sea goatfish line:
  - Begins at Winter, 2 fish
  - Rises gradually to Spring, 6 fish
  - Falls gradually to Fall, 0 fish
- The Milkfish line:
  - Begins at Winter, 12 fish
  - Falls gradually to Spring, 7 fish

- Falls gradually to Fall, 0 fish

Mangroves are trees or bushes that grow on the coastlines of seas and rivers. Areas with mangroves are great places for young fish since they help keep these fish fed and protected while they grow. To study the importance of mangroves to young fish, researchers Mohamed A.Abu El-Regal and Nesreen K. Ibrahim collected and identified young fish from three different mangrove sites in the Egyptian Red Sea. They collected fish in the winter, spring, and autumn of 2010, collecting a total of 269 fish from 21 different species. For some species, more fish were collected in the winter than the other two seasons, for instance: _____blank

Which choice most effectively uses the data in the graph to complete the example?

A. more common silver-biddy and milkfish were collected in the winter than in either of the other two seasons.

B. the common silver-biddy was collected more frequently than the other two species in all three seasons.

C. in the spring, researchers collected more Red Sea goldfish than they collected from the other two species.

D. in the fall, researchers collected 10 common silver-biddy but collected no milkfish or Red Sea goatfish.

2.6 d227c514-23b6-4498-90bf-633ef71950d4 · Medium

The Post Office is a 1912 play by Rabindranath Tagore, originally written in Bengali. The character Amal is a young boy who imagines that the people he sees passing the window of his home are carefree even when engaged in work or chores, as is evident when he says to the daughter of a flower seller, _____blank

Which quotation from *The Post Office* most effectively illustrates the claim?

A. “I see, you don’t wish to stop; I don’t care to stay on here either.”

B. “Oh, flower gathering? That is why your feet seem so glad and your anklets jingle so merrily as you walk.”

C. “I’ll pay when I grow up—before I leave to look for work out on the other side of that stream there.”

D. “Wish I could be out too. Then I would pick some flowers for you from the very topmost branches right out of sight.”

2.7 c26900f9-e2f8-4720-8940-4d2f422a2ee6 · Medium

Male túngara frogs make complex calls to attract mates, but their calls also attract frog-biting midges, insects that feed on the frogs' blood. Researchers Ximena Bernal and Priyanka de Silva wondered if the calls alone are sufficient for midges to locate the frogs or if midges use carbon dioxide emitted by frogs as an additional cue to their prey's whereabouts, like mosquitoes do. In an experiment, the researchers placed two midge traps in a túngara frog breeding area. One trap played recordings of túngara frog calls and the other released carbon dioxide along with playing the calls. Bernal and de Silva concluded that carbon dioxide does not serve as an additional cue to frog-biting midges.

Which finding from the experiment, if true, would most directly support Bernal and de Silva's conclusion?

A. Only a small number of midges were found in the traps, though the majority were found in the trap that played calls and released carbon dioxide.

B. Midges entered the trap that released carbon dioxide and played calls only during or immediately after periods of carbon dioxide release.

C. More midges were found in the trap that only played calls than in the trap that played calls and released carbon dioxide.

D. The trap that released carbon dioxide and played calls attracted few midges when carbon dioxide concentrations were low but attracted many midges when carbon dioxide concentrations were high.

2.8 29e2e0fc-8844-490c-85f3-24338f917e2a · Medium

Plants like potatoes, tomatoes, and soybeans are susceptible to bacterial wilt disease caused by the bacteria *Ralstonia solanacearum*. A multinational team of scientists led by Zhong Wei studied whether other microbes in the soil might influence the degree to which plants are affected by the disease. The team sampled soil surrounding individual tomato plants over time and compared the results of plants that became diseased with those that remained healthy. They concluded that the presence of certain microbes in the soil might explain the difference between healthy and diseased plants.

Which finding, if true, would most directly support the team's conclusion?

A. The soil surrounding healthy plants contained significantly higher concentrations of microbes known to inhibit *Ralstonia solanacearum* than the soil surrounding diseased plants did.

B. The soil surrounding the plants contained high concentrations of *Ralstonia solanacearum* regardless of whether the plants were affected by wilt disease.

C. The soil surrounding healthy plants tended to have significantly higher moisture levels than the soil surrounding diseased plants did.

D. By the end of the experiment, over half the plants had been affected by wilt disease regardless of differences in the types and concentrations of microbes in the surrounding soil.

2.9 56f276d0-7381-4372-865d-9872b5654569 · Medium

In the 1970s, a roughly 60,000-year-old piece of hyena bone marked with nine notches was discovered at a site in western France once inhabited by Neanderthals. Although many believe that only modern humans developed systems for notating numbers, one archaeologist asserts that this artifact may be a sign that Neanderthals also recorded numerical information. The notches on the bone are unevenly spaced but approximately parallel, and microscopic analysis reveals that they were made with a single stone tool; according to the archaeologist, this suggests that the notches were all made at one time by one individual as a means of counting something.

Which finding, if true, would most directly weaken the underlined claim?

A. Parallel lines are a common feature in modern humans' early systems for recording numerical information.

B. More than nine approximately parallel notches made with a different stone tool are present on another artifact found at a site in western France.

C. It would have taken careful effort to make evenly spaced lines on bone with the stone tools typically used by Neanderthals.

D. Decorative art discovered at another Neanderthal site in western France primarily features patterns of unevenly spaced parallel lines.

2.10 1a528880-397f-4739-b06d-5fa3d656aa3c · Medium

"Often Rebuked, Yet Always Back Returning" is an 1846 poem by Emily Brontë. The poem conveys the speaker's determination to experience the countryside around her: _____blank

Which quotation from the poem most effectively illustrates the claim?

A. "Often rebuked, yet always back returning / To those first feelings that were born with me, / And leaving busy chase of wealth and learning / For idle dreams of things which cannot be."

B. "I'll walk, but not in old heroic traces, / And not in paths of high morality, / And not among the half-distinguished faces, / The clouded forms of long-past history."

C. "I'll walk where my own nature would be leading: / It vexes me to choose another guide: / Where the grey flocks in ferny glens are feeding; / Where the wild wind blows on the mountain side."

D. "To-day, I will seek not the shadowy region; / Its unsustaining vastness waxes drear; / And visions rising, legion after legion, / Bring the unreal world too strangely near."

2.11 29d48243-c6fc-4b86-8b87-c1087174f7eb · Medium

A student is examining a long, challenging poem that was initially published in a quarterly journal without explanatory notes, then later republished in a stand-alone volume containing only that poem and accompanying explanatory notes written by the poet. The student asserts that the explanatory notes were included in the republication primarily as a marketing device to help sell the stand-alone volume.

Which statement, if true, would most directly support the student's claim?

A. The text of the poem as published in the quarterly journal is not identical to the text of the poem published in the stand-alone volume.

B. Many critics believe that the poet's explanatory notes remove certain ambiguities of the poem and make it less interesting as a result.

C. The publishers of the stand-alone volume requested the explanatory notes from the poet in order to make the book attractive to readers who already had a copy of the poem in a journal issue.

D. Correspondence between the poet and the publisher reveals that the poet's explanatory notes went through several drafts.

2.12 32c4c072-059d-4cea-9e8c-d27888cc1845 · Medium

"When Dawn Comes to the City" is a 1922 poem by Claude McKay, who immigrated to the United States from the island nation of Jamaica as an adult. The poem conveys McKay's contrasting feelings about New York City—his adopted home in the US—and his home country: _____blank

Which quotation from "When Dawn Comes to the City" most effectively illustrates the claim?

A. "A lonely newsboy hurries by, / Humming a recent ditty; / Red streaks strike through the gray of the sky, / The dawn comes to the city [New York City]."

B. "Dark figures start for work; / I watch them sadly shuffle on, / 'Tis dawn, dawn in New York. / But I would be on the island of the sea, / In the heart of the island of the sea."

C. "And the shaggy Nannie goat is calling, calling, calling / From her little trampled corner of the long wide lea / That stretches to the waters of the hill-stream falling / Sheer upon the flat rocks joyously!"

D. "The tired cars go grumbling by, / The moaning, groaning cars, / And the old milk carts go rumbling by / Under the same dull stars."

Effect of Paywall Introduction on Newspaper Companies' Revenues

Newspaper	Total revenue change ($ in thousands)	Percentage change (%)	Newspaper size
Los Angeles Times	93,966	12.5	large
The New York Times	235,788	20	large
The Denver Post	−3,765negative 3,765	−1negative 1	small
Sun Sentinel	−24,899negative 24,899	−11.9negative 11.9	small
Chicago Tribune	94,492	19	large

Digital paywalls restrict access to online content to those with a paid subscription. In an investigation of the effect of paywalls on newspaper company revenues for print and digital subscriptions and advertising, Doug J. Chung and colleagues compared actual outcomes (with a paywall) to control estimates (without a paywall). The researchers concluded that introducing a paywall is generally more beneficial for larger newspapers, which have high circulation and tend to offer a substantial amount of unique online content.

Which choice best describes data from the table that support Chung and colleagues' conclusion?

A. The *Chicago Tribune* and the *Los Angeles Times* had similar total revenue changes, but the *Los Angeles Times* had a smaller percentage change.

B. The *Los Angeles Times* had a 12.5% revenue change, while the *Chicago Tribune* had a 19% revenue change.

C. *The New York Times* had a 20% revenue change, while the *Denver Post* had a −1%negative 1% revenue change.

D. *The Denver Post* had only a −1%negative 1% revenue change, which was the smallest percentage change of the selected companies.

2.14 2882f061-3de5-48d0-8c0d-bca74bb2ec3f · Medium

“John of God, the Water-Carrier” is a 1913 short story by María Cristina Mena. In the story, the narrator presents John as being a hard worker who is fully dedicated to his job as water carrier, or *aguador*: _____blank

Which quotation from “John of God, the Water-Carrier” most effectively illustrates the claim?

A. “Very happy, he would jog home, the heavy silver pieces in his leather pockets making a discreet and dulcet ‘*trink-trak*’ between his jugs and his body.”

B. “He learned that the city *aguador* may not blow his whistle to halt the traffic while he gravely crosses the street, but must wait for the passing of many vehicles, some with horses and some outlandishly without.”

C. “From early morn to the fall of the afternoon he would go from fountain to fountain and from portal to portal, his lean body so accustomed to bending that he never thought of straightening it, his head bowed as if in prayer.”

D. “When his first jugs had worn out—the sweet-scented, porous red clay becomes perforated in time—he had buried them to their necks in the corner where he slept, and they were now his treasury.”

2.15 4722224d-0bd7-4a50-ae64-0069b6abcafc · Medium

Baltimore, Maryland, has installed engineered structures along 71% of its shoreline to protect infrastructure from wave erosion and other hazards, a practice known as shoreline hardening. To evaluate the responses of waterbirds to two types of hardening structures—riprap and bulkheads—Diann Prosser et al. surveyed waterbird communities consisting of the tundra swan, the great blue heron, and 62 other species at different sites in the Chesapeake Bay on the US East Coast. Utilizing the Index of Waterbird Community Integrity (IWCI), on which a high score corresponds to high community integrity, the researchers found that bulkheads are more strongly negatively correlated with waterbird community integrity than is riprap.

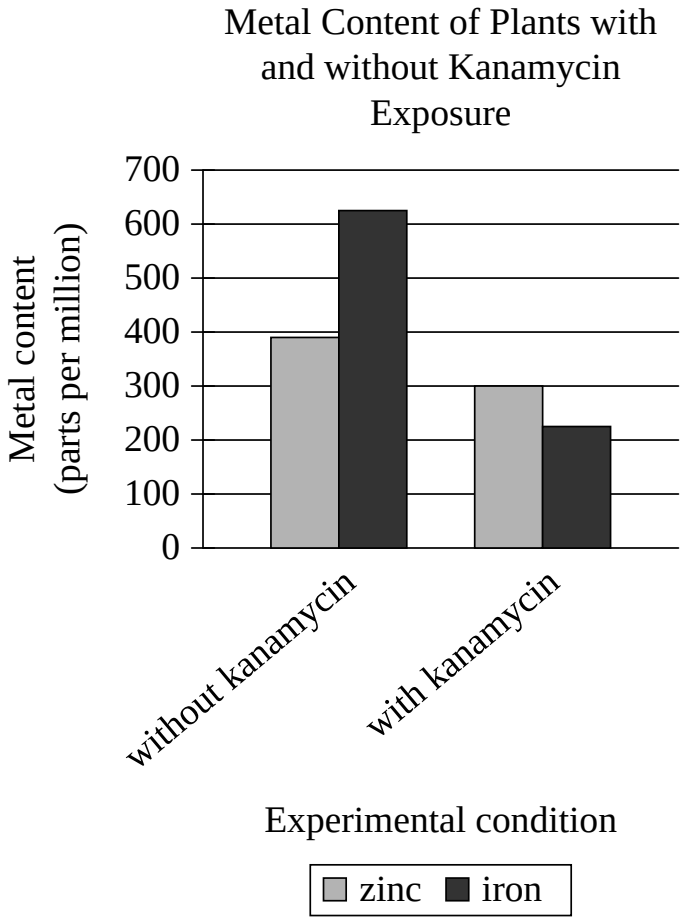
Which finding, if true, would most directly illustrate the researchers' finding?

A. The difference in average IWCI scores for waterbird communities at Stony and Old Road, two sites with a higher percentage of shoreline consisting of bulkheads than of riprap, was statistically insignificant.

B. Waterbird communities at Old Road, a site with a relatively high percentage of shoreline consisting of bulkheads, had lower average IWCI scores than did waterbird communities at Miles, a site with a relatively high percentage of shoreline consisting of riprap.

C. Waterbird communities at Curtis, a site with a high percentage of shoreline consisting of bulkheads and riprap, had lower average IWCI scores than did waterbird communities at Onancock, a site with a low percentage of shoreline consisting of bulkheads and riprap.

D. Waterbird communities at Curtis, a site with equal percentages of shoreline consisting of bulkheads and riprap, had higher average IWCI scores than did waterbird communities at Miles, a site with different percentages of shoreline consisting of bulkheads and riprap.



- For each data category, the following bars are shown:
  - Zinc
  - Iron
- The data for the 2 categories are as follows:
  - Without kanamycin:
    - Zinc: 390
    - Iron: 625
  - With kanamycin:
    - Zinc: 300
    - Iron: 225

Many plants lose their leaf color when exposed to kanamycin, an antibiotic produced by some soil microorganisms. Spelman College biologist Mentewab Ayalew and her colleagues hypothesized that plants' response to kanamycin exposure involves altering their uptake of metals, such as iron and zinc. The researchers grew two groups of seedlings of the plant *Arabidopsis thaliana*, half of

which were exposed to kanamycin and half of which were a control group without exposure to kanamycin, and measured the plants' metal content five days after germination.

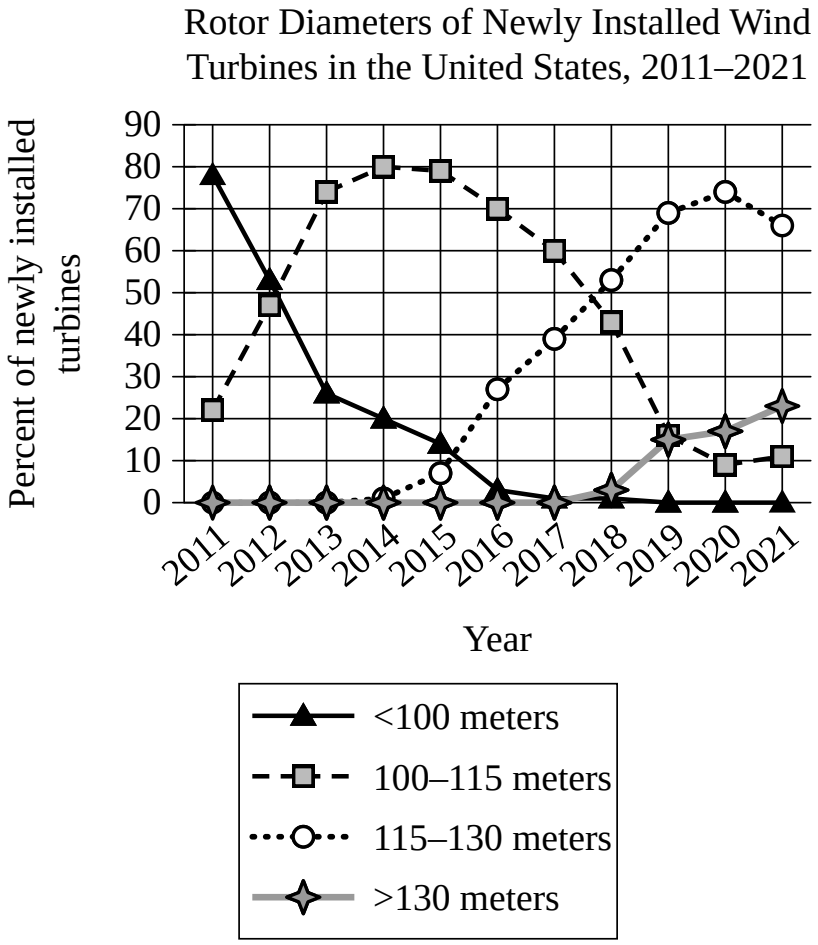
Which choice best describes data in the graph that support Ayalew and her colleagues' hypothesis?

A. The control plants contained higher levels of zinc than iron, but plants exposed to kanamycin contained higher levels of iron than zinc.

B. Both groups of plants contained more than 200 parts per million of both iron and zinc.

C. Zinc levels were around 300 parts per million in the control plants but nearly 400 parts per million in the plants exposed to kanamycin.

D. The plants exposed to kanamycin showed lower levels of iron and zinc than the control plants did.



All other things being equal, the larger a wind turbine's rotor diameter (the diameter of the imaginary circle swept by the turbine's rotating blades), the greater amount of energy the turbine can generate. In a research paper on wind power, a student claims that in the United States, the amount of energy generated per newly installed turbine increased substantially between 2011 and 2021.

Which choice best describes data in the graph that support the student's claim?

A. No turbines installed in 2011 had rotor diameters greater than 115 meters, whereas the majority of turbines installed in 2021 had rotor diameters greater than 130 meters.

B. Most turbines installed in 2011 had rotor diameters of less than 100 meters, whereas most turbines installed in 2021 had rotor diameters of at least 115 meters.

C. In 2011, nearly 80% of turbines installed had rotor diameters of less than 100 meters, whereas only a little more than 20% of turbines installed that year had rotor diameters of 100–115 meters.

D. The percentage of newly installed turbines with rotor diameters greater than 130 meters increased every year between 2011 and 2021.

2.18 5ae68e35-15b5-454d-9bc6-a20a205e4968 · Medium

In the early 1970s, art historian Michael Baxandall created an approach to viewing art called the "period eye," which explains how to look at art through the lens of its historical period. Baxandall argued that it is critical that art historians understand and communicate the original social and cultural contexts of older works of art so that it is clear what the artists intended and how the pieces would have been understood by their original viewers. Since it was first introduced, Baxandall's period eye has significantly influenced the practice of art history.

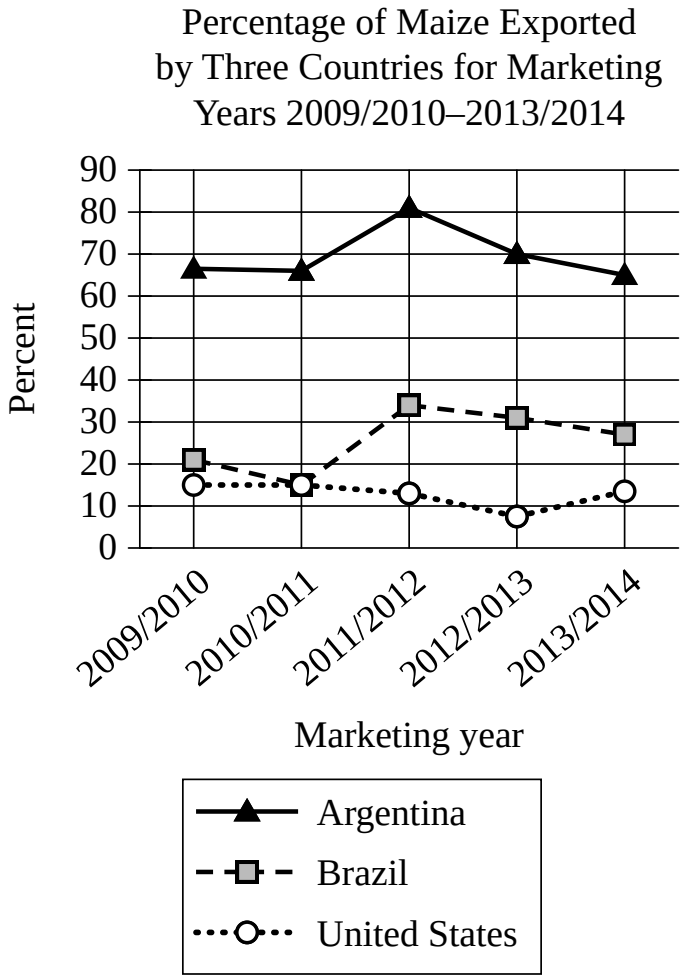
Which statement, if true, would most strongly support the claim in the underlined sentence?

A. Art historians working today have largely rejected the idea, common among Baxandall's predecessors, that artists' intentions should influence how artworks are interpreted.

B. For some historical periods, it is difficult for art historians to reconstruct how the original viewers of artworks understood what artists' intentions for their works were.

C. Numerous art historians of the late twentieth century and twenty-first century have focused their scholarship on how various artworks were interpreted at the time of their creation.

D. Many art historians working before the 1970s produced detailed analyses of the social and cultural contexts of older artworks, though few extended that approach to artworks produced in their own lifetimes.



- The following 3 lines are shown:
  - Argentina
  - Brazil
  - United States
- The Argentina line:
  - Begins at 2009/2010, 66.5%
  - Falls gradually to 2010/2011, 66%
  - Rises sharply to 2011/2012, 81%
  - Falls sharply to 2012/2013, 70%
  - Falls gradually to 2013/2014, 65%
- The Brazil line:
  - Begins at 2009/2010, 21%
  - Falls gradually to 2010/2011, 15%
  - Rises sharply to 2011/2012, 34%

    - Falls gradually to 2012/2013, 31%
    - Falls gradually to 2013/2014, 27%
- The United States line:
    - Begins at 2009/2010, 15%
    - Remains level to 2010/2011, 15%
    - Falls gradually to 2011/2012, 13%
    - Falls gradually to 2012/2013, 7.5%
    - Rises gradually to 2013/2014, 13.5%

Argentina, Brazil, and the United States are among the world's leading producers of maize (corn), and each country exports a certain percentage of maize each marketing year, which runs from March to February in Argentina and Brazil and from September to August in the United States. A student is researching those percentages and finds that for the marketing year 2012/2013, the percentage of maize exported by _____blank

Which choice most effectively uses data from the graph to complete the text?

A. Brazil increased from the previous marketing year but remained lower than the percentage exported by the United States.

B. Brazil exceeded the percentage exported by Argentina for the first time.

C. Argentina decreased from the previous marketing year but remained the highest among the three countries.

D. the United States reached its highest point during the five marketing years.

2.20 6b3e8eca-0f31-4ddc-ad58-4525b647b622 · Medium

“Mrs. Spring Fragrance” is a 1912 short story by Sui Sin Far. In the story, Mrs. Spring Fragrance, a Chinese immigrant living in Seattle, is traveling in California. In letters to her husband and friend, she demonstrates her concern for what’s happening at her home in Seattle while she is away: _____blank

Which quotation from Mrs. Spring Fragrance’s letters most effectively illustrates the claim?

A. “My honorable cousin is preparing for the Fifth Moon Festival, and wishes me to compound for the occasion some American ‘fudge,’ for which delectable sweet, made by my clumsy hands, you have sometimes shown a slight prejudice.”

B. “Next week I accompany Ah Oi to the beauteous town of San José. There will we be met by the son of the Illustrious Teacher.”

C. “Forget not to care for the cat, the birds, and the flowers. Do not eat too quickly nor fan too vigorously now that the weather is warming.”

D. “I am enjoying a most agreeable visit, and American friends, as also our own, strive benevolently for the accomplishment of my pleasure.”

2.21 34ff2bb7-fe6f-46f1-8559-4c79bc496112 · Medium

“On Virtue” is a 1766 poem by Phillis Wheatley. Wheatley addresses the poem directly to the quality of virtue, imploring it to assist her in reaching a future goal: _____blank

Which quotation from “On Virtue” most effectively illustrates the claim?

A. “Attend me, *Virtue*, thro’ my youthful years! / O leave me not to the false joys of time! / But guide my steps to endless life and bliss.”

B. “I cease to wonder, and no more attempt / Thine height t’explore, or fathom thy profound.”

C. “O thou bright jewel in my aim I strive / To comprehend thee. Thine own words declare / Wisdom is higher than a fool can reach.”

D. “But, O my soul, sink not into despair, / *Virtue* is near thee, and with gentle hand / Would now embrace thee, hovers o’er thine head.”

Five of the Responses to Survey about Actions to Conserve Energy

Action	Action category	Percentage of respondents selecting action (%)
Use efficient cars/hybrids	efficiency	2.8
Change thermostat setting	curtailment	6.3
Use bike or public transportation instead of car	curtailment	12.9
Use efficient light bulbs	efficiency	3.6
Turn off lights	curtailment	19.6

In a survey of public perceptions of energy use, researcher Shahzeen Attari and her team asked respondents to name the most effective action ordinary people can take to conserve energy. The team categorized each action as either an efficiency or a curtailment and found that respondents tended to name curtailments more often than they did efficiencies. For example, 19.6% of respondents stated that the most effective way to conserve energy is to turn off the lights, while only _____blank

Which choice most effectively uses data from the table to complete the text?

A. 6.3% of respondents said it was most effective to use efficient cars or hybrids.

B. 2.8% of respondents said it was most effective to change the thermostat setting.

C. 12.9% of respondents said it was most effective to use a bike or public transportation.

D. 3.6% of respondents said it was most effective to use efficient light bulbs.

2.23 a3f3d77e-57f2-4990-895d-58d6d0048f12 · Medium

“Ghosts of the Old Year” is an early 1900s poem by James Weldon Johnson. In the poem, the speaker describes experiencing an ongoing cycle of anticipation followed by regretful reflection: _____blank

Which quotation from “Ghosts of the Old Year” most effectively illustrates the claim?

A. “What does this brazen tongue declare, / That falling on the midnight air / Brings to my heart a sense of care / Akin to fright?”

B. “The snow has ceased its fluttering flight, / The wind sunk to a whisper light, / An ominous stillness fills the night, / A pause—a hush.”

C. “It tells of many a squandered day, / Of slighted gems and treasured clay, / Of precious stores not laid away, / Of fields unreaped.”

D. “And so the years go swiftly by, / Each, coming, brings ambitions high, / And each, departing, leaves a sigh / Linked to the past.”

2.24 a42ee406-d532-4af6-bce0-53b4aaf2ec41 · Medium

American fashion designer Patrick Kelly was known for his love of colorful buttons. Many of his signature dresses feature bold assortments of buttons throughout the garment. In a paper, a fashion design student claims that Kelly’s use of buttons as decoration was inspired by his childhood observations of the styles and actions of the women in his family.

Which quotation from a work by a historian would be the most effective evidence for the student to include in support of this claim?

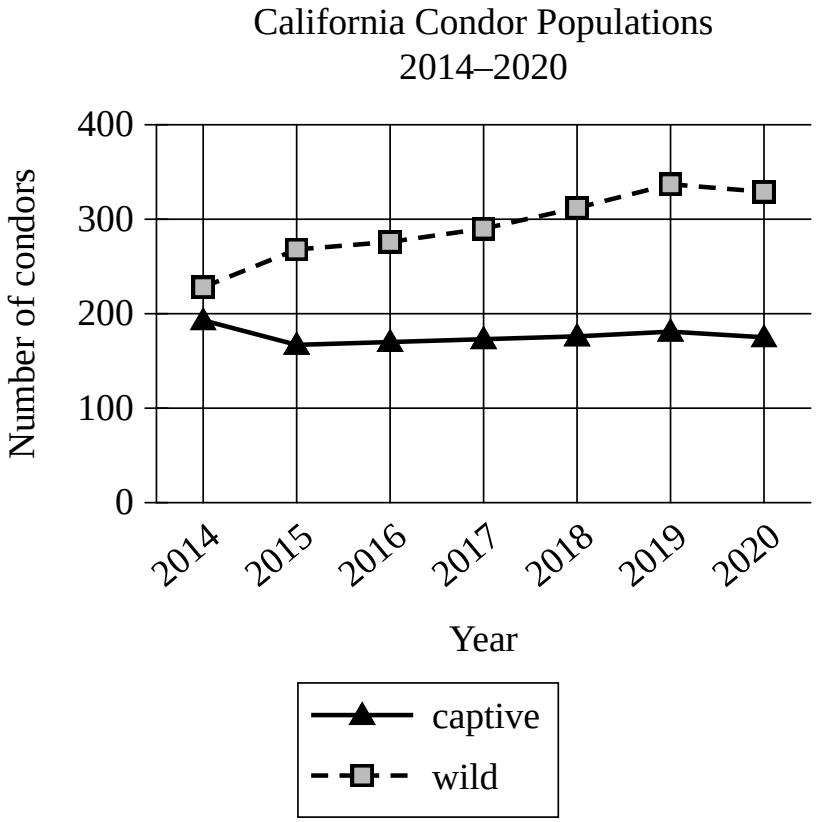
A. “Although some of the assortments of buttons appear to be mismatched pieces scattered randomly throughout Kelly’s dresses, his most famous designs feature carefully crafted patterns of matching buttons.”

B. “Many of Kelly’s contemporaries were inspired by his designs to incorporate buttons, as well as zippers and snaps, as decorative items in their work.”

C. “Kelly’s grandmother, who would repair clothing when he was a child, frequently added mismatched buttons to the clothes to draw attention away from any flaws in the garments.”

D. “Kelly was destined to be a designer from a young age: he learned how to sew clothing from his aunt Bertha, and his love of drawing was developed by his mother.”

2.25 368fc106-9c14-46b9-a363-0ebb5b9b5aea · Medium



- The following 2 lines are shown:
  - captive
  - wild
- The captive line:
  - Begins at 2014, 193 condors
  - Falls gradually to 2015, 167 condors
  - Rises gradually to 2016, 170 condors
  - Rises gradually to 2017, 173 condors
  - Rises gradually to 2018, 176 condors
  - Rises gradually to 2019, 181 condors
  - Falls gradually to 2020, 175 condors
- The wild line:
  - Begins at 2014, 228 condors
  - Rises gradually to 2015, 268 condors
  - Rises gradually to 2016, 276 condors
  - Rises gradually to 2017, 290 condors
  - Rises gradually to 2018, 312 condors
  - Rises gradually to 2019, 337 condors

- Falls gradually to 2020, 329 condors

The California Condor Recovery Program is working to protect this large, rare American vulture from extinction. The program captured 27 wild condors in 1987 in order to protect and breed them. Captive birds have been reintroduced into the wild since 1992. Although the California condor is still a rare species, many biologists claim that the program has been reasonably successful.

Which choice best describes data from the graph that support the biologists' claim?

A. The number of captive California condors steadily decreased between 2014 and 2020.

B. The numbers of California condors living both in captivity and in the wild decreased only slightly from 2019 to 2020.

C. The difference between the number of captive and wild California condors remained relatively steady between 2014 and 2017.

D. The number of wild California condors increased overall from 2014 to 2020 and exceeded the number of those living in captivity.

Properties of Select Rotating Radio Transients

Name	Right ascension (hours)	Period (seconds)	Frequency (hertz)
J0545-03	5:45	1.074	0.931
J1654-2335	16:54:03	0.545	1.834
J0103+54	1:03:37	0.354	2.822
J0121+53	1:21	2.725	0.367
J0614-03	6:15	0.136	7.353

A student is researching rotating radio transients (RRATs), a subclass of pulsar stars characterized by short pulses of radio waves. The time between consecutive pulses of an RRAT is referred to as a period. Looking at the table, the student determines that _____blank

Which choice most effectively uses data from the table to complete the statement?

A. J0614-03 has the shortest amount of time between consecutive pulses of all the RRATs in the table.

B. J0545-03 and J0121+53 have the same amount of time between consecutive pulses.

C. J1654-2335 has the longest amount of time between consecutive pulses of all the RRATs in the table.

D. J0103+54 and J0121+53 both have more than one second of time between consecutive pulses.

2.27 22a7e235-ee44-4c74-bca9-5b5d9a61b01a · Medium

Some residents in a neighborhood in Atlanta recently founded a community garden inside a local park. The residents agreed to volunteer to take care of the garden together. Students at a local high school surveyed some of the volunteers as part of a project to understand the impact of the new garden. The students concluded that the new garden benefited the community overall by fostering connections and relationships between the volunteers and other residents of the neighborhood who weren't volunteering at the garden.

Which quotation from a survey respondent would best illustrate the students' conclusion?

A. "Our first challenge was deciding what plants would be most suitable to the climate and soil here in Atlanta. We needed plants that could survive the hot and humid summers."

B. "We're lucky to have a few expert gardeners living in the neighborhood. Some volunteers and I have gone to them a few times with questions, and they've been eager to help us and to learn more about the project."

C. "I love getting the opportunity to be outside and around nature, especially on days when the weather is nice."

D. "My favorite thing about the garden is the feeling of pride I get when I walk by each day. As I see the plants growing, I feel good knowing I had a small part in creating this beautiful space in the neighborhood."

2.28 a358f59f-7a4a-4af5-a2a8-37a11ebbbec3 · Medium

"Looking Back on Girlhood" is an 1892 short story by Sarah Orne Jewett. In the story, the narrator explains that she prefers her hometown to other places she has visited: _____blank

Which quotation from "Looking Back on Girlhood" most effectively illustrates this claim?

A. "There is always something fresh, something to be traced or discovered, something particularly to be remembered."

B. "Two large rivers join just below the village at the head of tide-water, and these, with the great inflow from the sea, make a magnificent stream, bordered by lovely green fields that slope gently to long lines of willows at the water's edge."

C. "I have had a good deal of journeying in my life, and taken great delight in it, but I have never taken greater delight than in my rides and drives and tramps and voyages within the borders of my native town."

D. "There is never-ending pleasure in making one's self familiar with such a region."

2.29 4d6080a5-d5cb-4a7b-8d36-e063d820a389 · Medium

Roasted green chiles are a popular ingredient in Southwestern cuisine, but the traditional roasting method of burning propane is not environmentally friendly. To see if solar power could provide a better alternative, engineer Kenneth Armijo and his team roasted batches of green chiles using between 38 and 42 heliostats, which are devices that concentrate sunlight. The team was successful in reaching the same roasting temperature used in traditional propane roasting, but they found that propane yielded faster results. While the fastest solar-roasted green chiles took six minutes, batches using propane took only four. Armijo hypothesizes that they can reduce the roasting time for solar-roasted green chiles by using more heliostats.

Which finding, if true, would most directly support Armijo's hypothesis?

A. The temperature inside the roasting drum is distributed more evenly when roasting green chiles with solar power than with propane.

B. Attempts to roast green chiles using 50 heliostats yields results in fewer than six minutes.

C. Green chile connoisseurs prefer the flavor of solar-roasted green chiles over the flavor of propane-roasted green chiles.

D. The skins of solar-roasted green chiles are easier to peel than the skins of propane-roasted green chiles.

Approximate Rates of Speech and Information Conveyed for Five Languages

Language	Rate of speech (syllables per second)	Rate of information conveyed (bits per second)
Serbian	7.2	39.1
Spanish	7.7	42.0
Vietnamese	5.3	42.5
Thai	4.7	33.8
Hungarian	5.9	34.6

A group of researchers working in Europe, Asia, and Oceania conducted a study to determine how quickly different Eurasian languages are typically spoken (in syllables per second) and how much information they can effectively convey (in bits per second). They found that, although languages vary widely in the speed at which they are spoken, the amount of information languages can effectively convey tends to vary much less. Thus, they claim that two languages with very different spoken rates can nonetheless convey the same amount of information in a given amount of time.

Which choice best describes data from the table that support the researchers' claim?

A. Among the five languages in the table, Thai and Hungarian have the lowest rates of speech and the lowest rates of information conveyed.

B. Vietnamese conveys information at approximately the same rate as Spanish despite being spoken at a slower rate.

C. Among the five languages in the table, the language that is spoken the fastest is also the language that conveys information the fastest.

D. Serbian and Spanish are spoken at approximately the same rate, but Serbian conveys information faster than Spanish does.

Land Area Covered by Native Flowering Plants at a Site in Antarctica

Species	Area covered in 2009 (in square meters)	Area covered in 2018 (in square meters)	Percent increase in area covered from 2009 to 2018
Deschampsia antarctica	1,230	1,576	28%
Colobanthus quitensis	6.9	10.7	55%

The only flowering plant species native to Antarctica, *Colobanthus quitensis* and *Deschampsia antarctica* grow in places where the earth remains free of ice for much of the year. Botanist Niccoletta Cannone wondered how the warming of Antarctica's climate in recent years had affected these species, so she visited a site in Antarctica, first in 2009 and later in 2018, to count the number of plants growing there. Cannone found that the area of land covered by the two species had significantly expanded during the nine-year period. While both species likely benefited from warming temperatures, *Colobanthus quitensis* _____blank

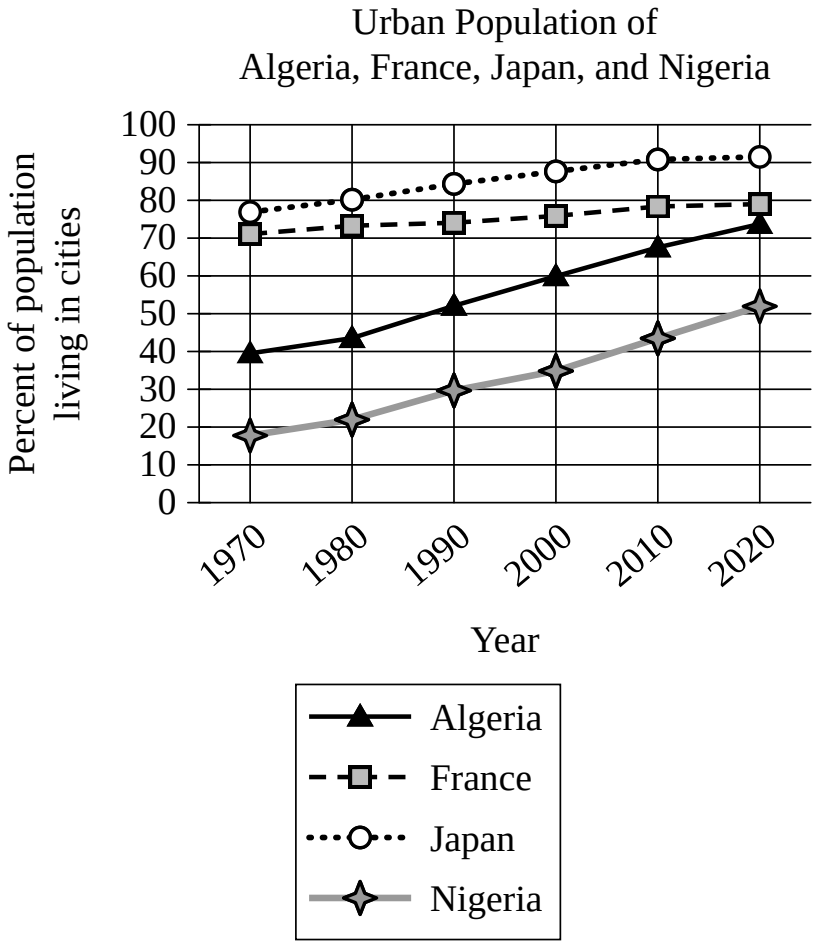
Which choice most effectively uses data from the table to complete the comparison?

A. suppressed the growth of *Deschampsia antarctica*, which covered a smaller area of land in 2018 than it had in 2009.

B. saw a greater expansion than *Deschampsia antarctica* did, increasing the area of land it covered by more than half.

C. showed a greater increase in the average size of individual plants than *Deschampsia antarctica* did.

D. covered land newly freed from ice at a rate 55% faster than that of *Deschampsia antarctica*.



- The following 4 lines are shown:
  - Algeria
  - France
  - Japan
  - Nigeria
- The Algeria line:
  - Begins at 1970, 39.5 percent
  - Rises gradually to 1980, 43.54 percent
  - Rises sharply to 1990, 52.09 percent
  - Rises gradually to 2000, 59.92 percent
  - Rises gradually to 2010, 67.54 percent
  - Rises gradually to 2020, 73.73 percent
- The France line:
  - Begins at 1970, 71.06 percent
  - Rises gradually to 1980, 73.28 percent

    - Rises gradually to 1990, 74.06 percent
    - Rises gradually to 2000, 75.87 percent
    - Rises gradually to 2010, 78.37 percent
    - Rises gradually to 2020, 79 percent
- The Japan line:
    - Begins at 1970, 76.88 percent
    - Rises gradually to 1980, 80.17 percent
    - Rises gradually to 1990, 84.34 percent
    - Rises gradually to 2000, 87.65 percent
    - Rises gradually to 2010, 90.81 percent
    - Rises gradually to 2020, 91.5 percent
- The Nigeria line:
    - Begins at 1970, 17.76 percent
    - Rises gradually to 1980, 21.97 percent
    - Rises gradually to 1990, 29.68 percent
    - Rises gradually to 2000, 34.84 percent
    - Rises sharply to 2010, 43.48 percent
    - Rises sharply to 2020, 51.96 percent

The share of the world's population living in cities has increased dramatically since 1970, but this change has not been uniform. France and Japan, for example, were already heavily urbanized in 1970, with 70% or more of the population living in cities. The main contributors to the world's urbanization since 1970 have been countries like Algeria, whose population went from _____blank

Which choice most effectively uses data from the graph to complete the assertion?

A. around 50% urban in 1970 to around 90% urban in 2020.

B. less than 40% urban in 1970 to around 90% urban in 2020.

C. less than 20% urban in 1970 to more than 50% urban in 2020.

D. around 40% urban in 1970 to more than 70% urban in 2020.

2.33 130d0a84-b1b0-4f4b-9201-cd5cc26e4514 · Medium

Sandra Cisneros's 1984 novella *The House on Mango Street* made a lasting impact on US literature. Its depiction of Mexican American culture inspired later authors to examine their own heritage within their fictional works. Also influential was the book's portrayal of the main character, Esperanza, during a pivotal year of her youth. This insightful depiction of a preteen girl encouraged authors who, like Cisneros herself, are Latina to use fictional works to examine experiences from their own youth.

Which statement, if true, would most strongly support the claim in the underlined sentence?

A. In interviews, a number of Latina authors say that *The House on Mango Street* inspired them to write about their own adolescence in their novels.

B. In published writings, several prominent authors who are not Latina say that reading *The House on Mango Street* influenced their approach to writing fiction.

C. The *House on Mango Street* has sold over six million copies and is one of the most commonly read books among high school and university students in the US.

D. Since 1984, new novels about young Latina characters by Latina authors have often been compared to *The House on Mango Street*.

Mean Ratings for Patients after 21 Days

Measure	Mean rating for participants aware of taking a placebo	Mean rating for participants in the control group
Global improvement	5.0	3.9
Symptom severity reduction	92.00	46.00
Quality of life improvement	11.4	5.4

To test whether a medication is effective, scientists compare outcomes for patients taking it and patients taking a placebo (a medically inactive substance). Patients normally aren't told they're receiving a placebo, but a research team conducted a study to see if there might be a medical benefit to telling them. The team used various measures to evaluate participants, with higher ratings indicating greater well-being in each measure. Compared to the mean ratings after 21 days for participants in the control group, the mean ratings for participants who were aware of taking a placebo _____blank

Which choice most effectively uses data from the table to complete the statement?

A. ranged from 5.0 to 92.00, indicating that well-being varied widely from participant to participant.

B. were lower for two measures, with the rating for only one measure indicating greater well-being for these participants.

C. ranged from 3.9 to 46.00, with no rating indicating greater well-being in any measure for these participants.

D. were higher for all three measures, indicating greater overall well-being for these participants.

2.35 f18119e1-0678-452f-bee5-0dfaddbe173f · Medium

Annual Spending by International Tourists in Four Countries (in billions of US dollars)

Country	2016	2017	2018
South Korea	$21.0	$17.2	$23.1
Japan	$33.5	$37.0	$45.3
Thailand	$48.5	$57.1	$61.4
Malaysia	$19.7	$20.3	$21.8

One measure of international tourism is how much money visitors from abroad spend in a country. A student writing a report about tourism in several Asian countries notes that among the countries in the table, most had annual increases in international tourism revenue in 2017 and 2018, but that trend wasn't universal. For example, _____blank

Which choice most effectively uses data from the table to complete the example?

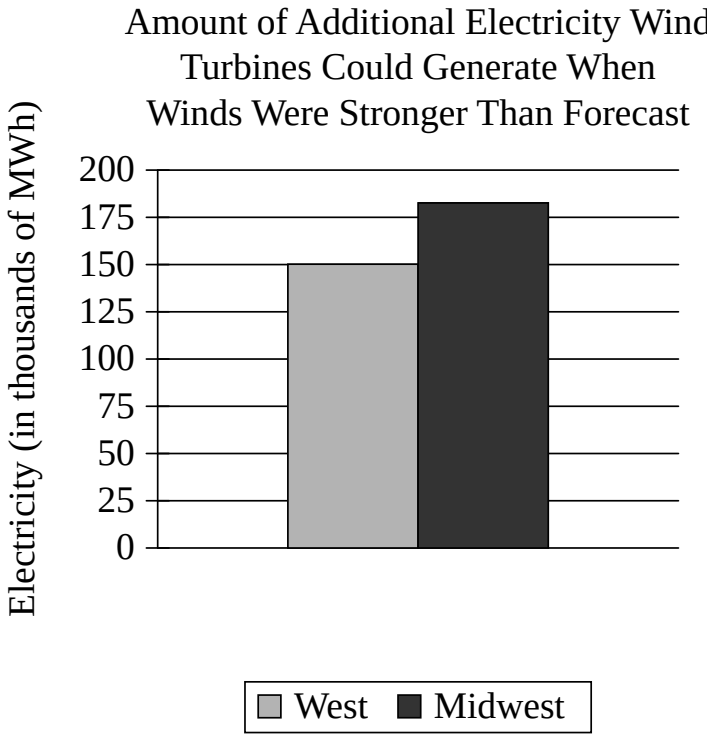
A. in each year shown, the revenue from tourism by residents of those countries was greater than the revenue from international tourism.

B. Japan's international tourism revenue was higher in 2016 than it was in 2018.

C. South Korea's annual international tourism revenue decreased from 2016 to 2017.

D. in each year shown, Thailand had higher international tourism income than any of the other countries.

2.36 2b5d289b-aa4a-455f-9d3c-3fb0079cac56 · Medium



- The data for the 2 categories are as follows:
  - West: 150.2 thousand megawatt-hours
  - Midwest: 182.6 thousand megawatt-hours

Electric companies that use wind turbines rely on weather forecasts to predict the maximum amount of power, in megawatt-hours (MWh), they can generate using wind so that they can determine how much they'll need to generate from other sources. When winds are stronger than they were forecast to be, however, the predicted maximum amount of electricity wind turbines could generate will be too low. For example, the graph shows that for the West region, the winds were _____blank

Which choice most effectively uses data from the graph to complete the example?

A. strong enough to generate about 150 thousand more MWh of electricity from wind turbines.

B. so weak that the electricity from wind turbines was about 175 thousand MWh less than predicted.

C. so weak that the electricity from wind turbines was about 150 thousand MWh less than predicted.

D. strong enough to generate about 175 thousand more MWh of electricity from wind turbines.

2.37 1d6e68b4-e317-4866-ac32-947f85936b85 · Medium

East Australian humpback whales migrate up to 10,000 kilometers each year to reach their breeding grounds. Researchers long believed that migrating whales live only on the extra energy they stored up during the feeding season. But marine biologist Vanessa Pirotta and her team aren't so sure. They analyzed 20 years of observations of the migrating whales made by citizen scientists (members of the public who help collect data for scientific research). The team claims that the whales may not live only on their stored energy during migration.

Which finding, if true, would most directly support the team's claim?

A. Citizen scientists have observed many different types of marine animals feeding alongside the whales.

B. Citizen scientists have made many observations of the whales feeding as they migrate to their breeding grounds.

C. Citizen scientists have made more observations of the whales migrating to their breeding grounds than of the whales returning to their feeding grounds.

D. Citizen scientists have recently begun to observe the whales migrating to their breeding grounds earlier in the year.

2.38 2b554656-dea7-4696-877f-cfcf47558291 · Medium

Studies have demonstrated that positive feedback enhances real-world exercise performance and exercisers' psychological experience of physical activity. Nicole Trewick and team tested their prediction that positive feedback would produce analogous results among participants cycling on a stationary bike in a virtual reality environment. After monitoring participants' pedaling rate and heart rate to determine the effects on their physical endurance (e.g., their ability to sustain a certain target pedaling and heart rate) of feedback (positive, negative, or neutral) delivered at regular intervals, the researchers used participants' questionnaire responses to assess their psychological experience of the task.

Assuming participants had similar baseline fitness levels, which finding from the study, if true, would most strongly suggest that positive feedback had the predicted psychological effect but not the predicted physical effect?

A. Compared with participants who received positive feedback, participants who received neutral feedback reported lower enjoyment of the activity on average but maintained their heart and pedaling rates for longer durations than participants who received negative feedback.

B. Compared with participants who received positive or neutral feedback, participants who received negative feedback reported lower enjoyment of the activity on average but maintained their heart and pedaling rates for longer durations.

C. Compared with participants who received negative feedback, participants who received neutral feedback reported similar levels of enjoyment of the activity on average but maintained their heart and pedaling rates for shorter durations than participants who received positive feedback.

D. Compared with participants who received negative or neutral feedback, participants who received positive feedback reported greater enjoyment of the activity on average but maintained their heart and pedaling rates for approximately similar durations.

2.39 ad1536b8-b043-4d8d-8946-9aef5ae9a0fc · Medium

In 1967 the US Congress created the Corporation for Public Broadcasting, which in turn created National Public Radio (NPR). NPR began producing and distributing high-quality news and cultural programming to affiliate stations across the United States in 1971. In a research paper, a student claims that the Corporation for Public Broadcasting and NPR were inspired by the British Broadcasting System (BBC), which had been established in the 1920s.

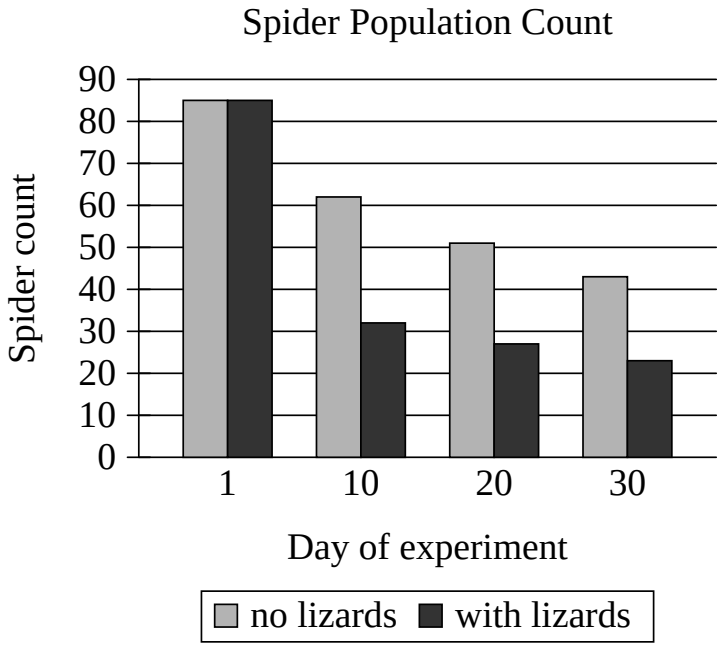
Which quotation from a work by a historian would be the most effective evidence for the student to include in support of this claim?

A. “Although the BBC had begun as a private corporation, politicians successfully argued to make it a public company because they believed a public broadcaster could help build national unity in the aftermath of World War I.”

B. “For many decades, the BBC had no competition since it held Britain’s only broadcasting license, whereas in the United States, the Corporation for Public Broadcasting launched NPR in a broadcasting market already filled with competitors.”

C. “Congress’s embrace of publicly funded broadcasting reflected a common belief among US politicians that the role of government was not only to ensure people’s safety and liberty but also to enrich people’s lives in other ways.”

D. “The goal of the BBC was to support British democracy by promoting an informed citizenry, and US legislators believed that ensuring access to high-quality programming could do the same for democracy in the United States.”



- For each data category, the following bars are shown:
  - no lizards
  - with lizards
- All values are approximate.
- The data for the 4 categories are as follows:
  - Day 1:
    - no lizards: 85
    - with lizards: 85
  - Day 10:
    - no lizards: 62
    - with lizards: 32
  - Day 20:
    - no lizards: 51
    - with lizards: 27
  - Day 30:
    - no lizards: 43
    - with lizards: 23

To investigate the effect of lizard predation on spider populations, a student in a biology class placed spiders in two enclosures, one with lizards and one without, and tracked the number of spiders in the enclosures for 30 days. The student concluded that the reduction in the spider

population count in the enclosure with lizards by day 30 was entirely attributable to the presence of the lizards.

Which choice best describes data from the graph that weaken the student's conclusion?

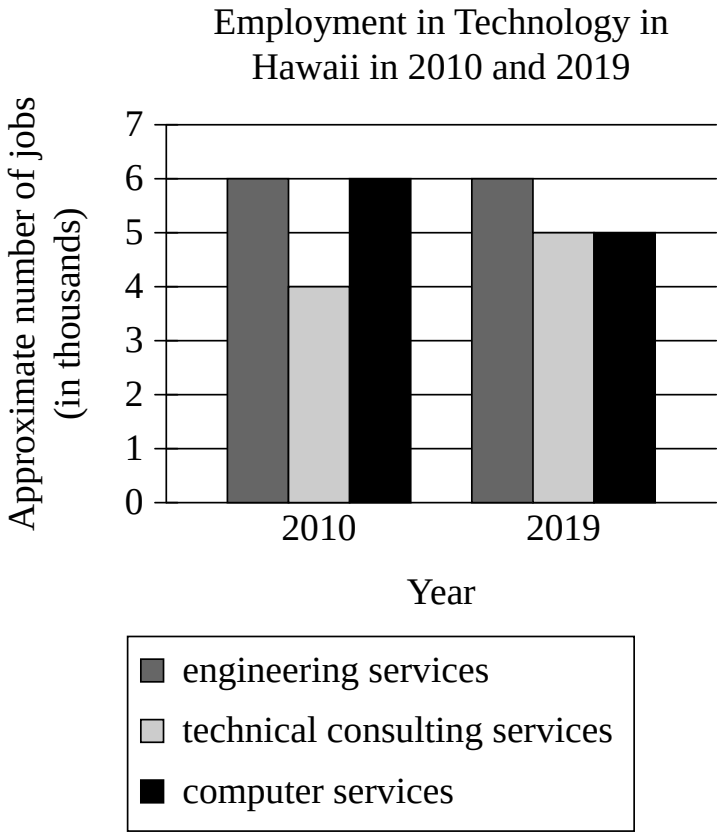
A. The spider population count was the same in both enclosures on day 1.

B. The spider population count also substantially declined by day 30 in the enclosure without lizards.

C. The largest decline in spider population count in the enclosure with lizards occurred from day 1 to day 10.

D. The spider population count on day 30 was lower in the enclosure with lizards than in the enclosure without lizards.

2.41 4d0eef89-bee9-47f6-ad14-52c8d8b98ced · Medium



- For each data category, the following bars are shown:
  - engineering services
  - technical consulting services
  - computer services
- All data are approximate.
- The Approximate number of jobs data for the 2 categories are as follows:
  - 2010:
    - engineering services: 6 thousand jobs
    - technical consulting services: 4 thousand jobs
    - computer services: 6 thousand jobs
  - 2019:
    - engineering services: 6 thousand jobs
    - technical consulting services: 5 thousand jobs
    - computer services: 5 thousand jobs

A student in Hawaii is interested in pursuing a career in technology and decides to do some research on local trends. The student notices that the number of jobs in computer services in 2010 was _____blank

Which choice most effectively uses data from the graph to complete the statement?

A. higher than the number of jobs in technical consulting services, and in 2019 was about the same as the number of jobs in engineering services.

B. about the same as the number of jobs in engineering services, and in 2019 was about the same as the number of jobs in technical consulting services.

C. lower than the number of jobs in engineering services, but in 2019 was higher than the number of jobs in engineering services.

D. about the same as the number of jobs in technical consulting services, but in 2019 was lower than the number of jobs in technical consulting services.

2.42 dbd13420-a4fe-4004-886a-8f3338082300 · Medium

“Odalie” is an 1899 short story by Alice Dunbar-Nelson. In the story, a young woman named Odalie attends the annual Mardi Gras carnival in New Orleans, where she lives with her guardian Tante Louise. Dunbar-Nelson portrays Odalie as eager to escape the monotony of her everyday life: _____blank

Which quotation from “Odalie” most effectively illustrates the claim?

A. “Mardi Gras was a tiresome day, after all, she sighed, and Tante Louise agreed with her for once.”

B. “In the old French house on Royal Street, with its quaint windows and Spanish courtyard green and cool, and made musical by the plashing of the fountain and the trill of caged birds, lived Odalie in convent-like seclusion.”

C. “When one is shut up in a great French house with a grim sleepy tante and no companions of one’s own age, life becomes a dull thing, and one is ready for any new sensation.”

D. “It was Mardi Gras day at last, and early through her window Odalie could hear the jingle of folly bells on the [participants’] costumes, the tinkle of music, and the echoing strains of songs.”

Body Length, Filter Time, and Lunges per Dive for Four Whale Species

Whale species	Typical adult body length (meters)	Average time to filter all engulfed water (seconds)	Average number of lunges per dive deeper than 50 meters
fin	18–22	31.30	3.95
humpback	11–17	17.12	6.28
minke	7–10	8.88	7.48
blue	24–34	60.27	4.02

Some whale species practice lunge feeding, in which they lunge toward prey with their mouths open at wide angles, collect the prey and the surrounding water, and then filter out the water through baleen plates in their mouths. Although the volume of water engulfed increases with whales' body length, the surface area of whales' baleen plates, which influences the rate at which water can be filtered, does not increase with body length to the same degree, which helps explain why _____blank

Which choice most effectively uses data from the table to complete the statement?

A. minke whales and humpback whales show similar average filter times.

B. humpback whales show an average of 6.28 lunges per dive.

C. fin whales show a longer average filter time than minke whales do.

D. blue whales show the longest average filter time and the highest average number of lunges per dive.

E-book Sales as a Percentage of Total Unit Sales in All Book Formats for a Large US Trade Publisher, by Genre, 2006, 2011, 2016

Genre	2006	2011	2016
science fiction and fantasy	0.6	27.7	36.7
cookbooks	0	2.9	10.5
travel guides	0	5.5	24.6
romance	0.3	40.6	56.2

E-books became an increasingly popular means of reading in the United States in the 2000s and 2010s, though that popularity was concentrated in titles that, like those in most fiction genres, are meant to be read straight through from beginning to end. For books in nonfiction genres that do not tell stories and require the reader to flip back and forth through a volume, e-books were significantly less commercially successful. This can be seen by comparing _____blank

Which choice most effectively uses data from the table to illustrate the claim?

A. the percentage of 2016 cookbook sales that were e-books with the percentage of 2016 science fiction and fantasy sales that were e-books.

B. the percentage of 2006 romance sales that were e-books with the percentage of 2016 romance sales that were e-books.

C. the percentage of 2006 romance sales that were e-books with the 2006 science fiction and fantasy sales that were e-books.

D. the percentage of 2011 travel guide sales that were e-books with the percentage of 2016 travel guide sales that were e-books.

Dated Ages of Lunar Samples from Select Missions

Mission name	Year	Landing site	Approximate age of lunar samples (billions of years)
Apollo 11	1969	Mare Tranquillitatis	3.6
Apollo 15	1971	Mare Imbrium	3.3
Apollo 17	1972	Mare Serenitatis	3.8
Chang'e 5	2020	Oceanus Procellarum	2.0

The Apollo program missions were spaceflights to the moon led by the United States during the 1960s and 1970s during which astronauts collected some samples of the moon's surface. More recently, China launched the Chang'e 5 mission, which returned additional lunar surface samples. Researchers have analyzed and dated each of the samples, concluding that the lunar samples collected during the Chang'e 5 mission are significant because _____blank

Which choice most effectively uses data from the table to complete the claim?

A. they are much younger than the samples brought back from any of the Apollo missions.

B. they were collected from the same landing site as the Apollo 11 mission.

C. they are closest in age to the samples brought back by the Apollo 17 mission.

D. they helped confirm the predicted ages of the lunar samples from the Apollo missions.

Characteristics of Five Recently Discovered Gas Exoplanets

Exoplanet designation	Mass (Jupiters)	Radius (Jupiters)	Orbital period (days)	Distance from the Sun (parsecs)
TOI-640 b	0.88	1.771	5.003	340
TOI-1601 b	0.99	1.239	5.331	336
TOI-628 b	6.33	1.060	3.409	178
TOI-1478 b	0.85	1.060	10.180	153
TOI-1333 b	2.37	1.396	4.720	200

“Hot Jupiters” are gas planets that have a mass of at least 0.25 Jupiters (meaning that their mass is at least 25% of that of Jupiter) and an orbital period of less than 10 days (meaning that they complete one orbit around their star in less than 10 days), while “warm Jupiters” are gas planets that meet the same mass criterion but have orbital periods of more than 10 days. In 2021, Michigan State University astronomer Joseph Rodriguez and colleagues announced the discovery of five new gas exoplanets and asserted that four are hot Jupiters and one is a warm Jupiter.

Which choice best describes data from the table that support Rodriguez and colleagues’ assertion?

A. None of the planets have an orbital period of more than 10 days, and TOI-628 b has a mass of 6.33 Jupiters.

B. TOI-1478 b has an orbital period of 153 days, and the masses of all the planets range from 0.85 to 6.33 Jupiters.

C. All the planets have a radius between 1.060 and 1.771 Jupiters, and only TOI-1333 b has an orbital period of more than 10 days.

D. Each of the planets has a mass greater than 0.25 Jupiters, and all except for TOI-1478 b have an orbital period of less than 10 days.

2.47 2d43925d-e0c3-44ec-b583-82c5c0d2a74e · Medium

“The Yellow Wallpaper” is an 1892 short story by Charlotte Perkins Gilman. In the story, the narrator expresses mixed feelings about her surroundings: _____blank

Which quotation from “The Yellow Wallpaper” most effectively illustrates the claim?

A. “This wallpaper has a kind of sub-pattern in a different shade, a particularly irritating one, for you can only see it in certain lights, and not clearly then.”

B. “By moonlight—the moon shines in all night when there is a moon—I wouldn’t know it was the same paper.”

C. “I’m really getting quite fond of the big room, all but that horrid [wall]paper.”

D. “The color is repellant, almost revolting; a smouldering, unclean yellow, strangely faded by the slow-turning sunlight.”

2.48 e937bcae-33f9-41e7-a260-badb93557071 · Medium

Many music streaming services also function as social media platforms: by giving users the option to follow friends and share curated playlists or information about their listening history, these platforms allow people to convey their music preferences and listening activities directly to their social networks. In a 2023 study, researcher Michael James Walsh interviewed frequent users of a popular music streaming platform to investigate how its social media dimensions shape their listening habits. Walsh found that these dimensions tended to make study participants feel more mindful of how their listening activities may be perceived, which in turn influenced how they managed those activities.

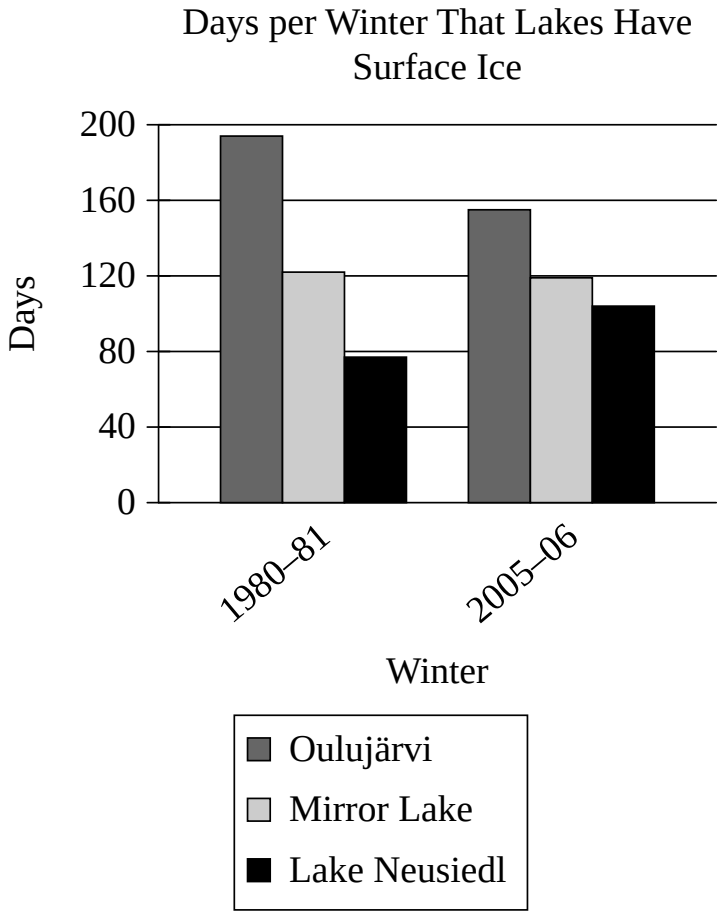
Which quotation from a music streaming platform user would best illustrate Walsh’s finding?

A. “When I create playlists or choose songs to listen to, I often pick songs that I think the friends who are following me might be interested in listening to, rather than what I might actually prefer listening to in that moment.”

B. “Because the predictions the platform makes about what new songs I might like depend on my past listening habits, at times its recommendations can feel overly repetitive and unvarying.”

C. “The social aspects of music streaming make it really easy to create a sense of a shared listening experience with my friends, even if we aren’t in the same room.”

D. “Listening to music through a streaming platform feels a lot more passive, since I don’t need to make any deliberate choices about what music I’m going to consume, like I do when buying digital downloads or records and CDs.”



- For each data category, the following bars are shown:
  - Oulujärvi
  - Mirror Lake
  - Lake Neusiedl
- The data for the 2 categories are as follows:
  - 1980–81:
    - Oulujärvi: 194 days
    - Mirror Lake: 122 days
    - Lake Neusiedl: 77 days
  - 2005–06:
    - Oulujärvi: 155 days
    - Mirror Lake: 119 days
    - Lake Neusiedl: 104 days

It is common for freshwater lakes near or above a latitude of 45° north of the equator, like Lake Mjøsa in Norway, to accumulate surface ice in winter. The amount and duration of ice depends on

many factors, including local weather conditions as well as the lake's depth, volume, and surface area, but a climate researcher claims that some lakes in these latitudes have seen a decline in the duration of ice between the early 1980s and the mid-2000s. She cites as a typical example _____blank

Which choice most effectively uses data from the graph to complete the researcher's example?

A. both Lake Neusiedl and Oulujärvi, which had fewer than 195 days of ice in the winter of 1980–81.

B. Lake Neusiedl, which had more days of ice in the winter of 2005–06 than it did in the winter of 1980–81.

C. Oulujärvi, which had fewer days of ice in the winter of 2005–06 than it did in the winter of 1980–81.

D. both Lake Neusiedl and Oulujärvi, which had more than 105 days of ice in the winter of 2005–06.

2.50 33334091-eb57-44af-95a6-fbc105389a91 · Medium

Puerto Rico is an island in the Caribbean Sea. Indigenous people there started raising guinea pigs about 1,700 years ago. Guinea pigs had originally been domesticated much earlier in both Colombia and Peru. So were guinea pigs brought to Puerto Rico from Colombia or from Peru? Ancient Caribbean trade routes connected Puerto Rico with Colombia but not with Peru. Therefore, guinea pigs in Puerto Rico probably came from Colombia and descended from Colombian guinea pigs.

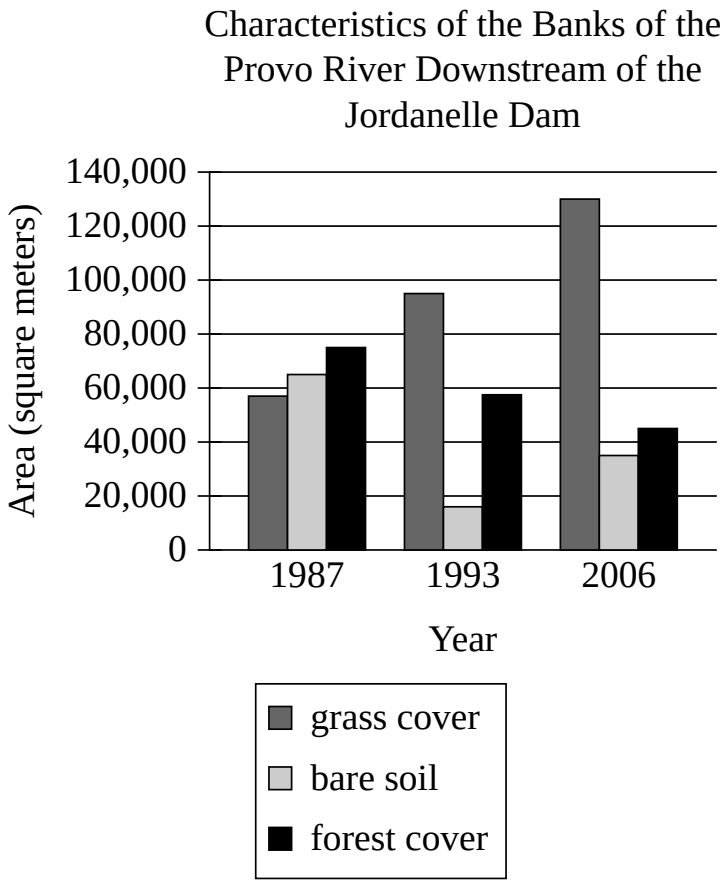
Which finding, if true, would most directly weaken the underlined claim?

A. Ancient guinea pigs in Puerto Rico were genetically less similar to ancient guinea pigs in Colombia than to ancient guinea pigs in Peru.

B. Guinea pigs are common in ancient Puerto Rican art, especially in pottery.

C. Modern breeds of guinea pigs don't look like images of guinea pigs in ancient art from Puerto Rico, Colombia, and Peru.

D. The guinea pig population of ancient Colombia was much larger than the guinea pig population of ancient Peru.



- For each data category, the following bars are shown:
  - grass cover
  - bare soil
  - forest cover
- All values are approximate.
- The data for the 3 categories are as follows:
  - 1987:
    - grass cover: 57,000 square meters
    - bare soil: 65,000 square meters
    - forest cover: 75,000 square meters
  - 1993:
    - grass cover: 95,000 square meters
    - bare soil: 16,000 square meters
    - forest cover: 57,500 square meters
  - 2006:
    - grass cover: 130,000 square meters

- bare soil: 35,000 square meters
- forest cover: 45,000 square meters

The Jordanelle Dam was built on the Provo River in Utah in 1992. Earth scientist Adriana E. Martinez and colleagues tracked changes to the environment on the banks of the river downstream of the dam, including how much grass and forest cover were present. They concluded that the dam changed the flow of the river in ways that benefited grass plants but didn't benefit trees.

Which choice best describes data from the graph that support Martinez and colleagues' conclusion?

A. The lowest amount of grass cover was approximately 58,000 square meters, and the highest amount of forest cover was approximately 75,000 square meters.

B. There was more grass cover than forest cover in 1987, and this difference increased dramatically in 1993 and again in 2006.

C. There was less grass cover than bare soil in 1987 but more grass cover than bare soil in 1993 and 2006, whereas there was more forest cover than bare soil in all three years.

D. Grass cover increased from 1987 to 1993 and from 1993 to 2006, whereas forest cover decreased in those periods.

2.52 120d4012-6089-4617-a7f4-e27c49c875e1 · Medium

Although most songbirds build open, cupped nests, some species build domed nests with roofs that provide much more protection. Many ecologists have assumed that domed nests would provide protection from weather conditions and thus would allow species that build them to have larger geographic ranges than species that build open nests do. To evaluate this assumption, a research team led by evolutionary biologist Iliana Medina analyzed data for over 3,000 species of songbirds.

Which finding from Medina and her colleagues' study, if true, would most directly challenge the assumption in the underlined sentence?

A. Species that build open nests tend to have higher extinction rates than species that build domed nests.

B. Species that build open nests tend to be smaller in size than species that build domed nests.

C. Species that build open nests tend to use fewer materials to build their nests than species that build domed nests do.

D. Species that build open nests tend to have larger ranges than species that build domed nests.

Participants' Evaluation of the Likelihood That Robots Can Work Effectively in Different Occupations

Occupation	Somewhat or very unlikely (%)	Neutral (%)	Somewhat or very likely (%)
television news anchor	24	9	67
teacher	37	16	47
firefighter	62	9	30
surgeon	74	9	16
tour guide	10	8	82
Rows in table may not add up to 100 due to rounding.			

Georgia Tech roboticists De'Aira Bryant and Ayanna Howard, along with ethicist Jason Borenstein, were interested in people's perceptions of robots' competence. They recruited participants and asked them how likely they think it is that a robot could do the work required in various occupations. Participants' evaluations varied widely depending on which occupation was being considered; for example, _____blank

Which choice most effectively uses data from the table to complete the example?

A. 47% of participants believe that it is somewhat or very likely that a robot could work effectively as a teacher, but 37% of respondents believe that it is somewhat or very unlikely that a robot could do so.

B. 9% of participants were neutral about whether a robot could work effectively as a television news anchor, which is the same percent of participants who were neutral when asked about a robot working as a surgeon.

C. 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide, but only 16% believe that it is somewhat or very likely that a robot could work as a surgeon.

D. 62% of participants believe that it is somewhat or very unlikely that a robot could work effectively as a firefighter.

2.54 a0624880-0cff-4f54-93c7-0ee67780fa3a · Medium

Rivers rich in sediment appear yellow, while increases in red algae make rivers appear red. To track things like the sediment or algae content of large US rivers, John R. Gardner and colleagues used satellite data to determine the dominant visible wavelengths of light measured for various segments of these rivers. The researchers classified wavelengths of 495 nanometers (nm) and below as red, wavelengths between 495 and 560 nm as blue, and wavelengths of 560 nm and above as yellow. The researchers concluded that for the Missouri River, segments flowing into lakes tend to carry more sediment than those flowing out of lakes.

Which finding, if true, would most directly support the researchers' conclusion?

A. The segments of the Missouri River that had higher levels of chlorophyll-a, which contributes to the green color of photosynthetic organisms, have dominant wavelengths of light between 490 and 560 nm.

B. In lakes through which segments of the Missouri River pass, the dominant wavelength of light tended to be above 560 nm near the lakes' shores and below 560 nm in the lakes' centers.

C. The majority of the segments of the Missouri River were found to have dominant wavelengths of light significantly higher than 560 nm.

D. Segments of the Missouri River flowing into lakes typically had dominant wavelengths of light above 560 nm, while segments flowing out of lakes typically had dominant wavelengths below 560 nm.

Partial List of Candidate Species for De-extinction

Common name	Scientific name	Became extinct
Huia	*Heteralocha acutirostris*	1907
Caribbean monk seal	*Monachus tropicalis*	1952
Passenger pigeon	*Ectopistes migratorius*	1914
Saber-toothed cat	*Smilodon*	11,000 years before present
Woolly mammoth	*Mammuthus primigenius*	6,400 years before present

The passage of time is among the many obstacles faced by scientists who are pursuing de-extinction efforts—that is, efforts to use breeding or a mixture of cloning and genetic engineering to bring back extinct species. Specifically, researchers are concerned that the longer a species has been extinct, the less likely it is that a suitable habitat still exists for that species. Among candidate species for de-extinction, this problem would be especially concerning for the _____blank

Which choice most effectively uses data from the table to complete the statement?

A. passenger pigeon (*Ectopistes migratorius*), which became extinct only a few years after the huia (*Heteralocha acutirostris*).

B. saber-toothed cat (*Smilodon*), which became extinct 11,000 years ago.

C. woolly mammoth (*Mammuthus primigenius*), which became extinct several thousand years before the saber-toothed cat (*Smilodon*).

D. Caribbean monk seal (*Monachus tropicalis*), which became extinct in 1952.

2.56 3cc18ae3-a388-4035-8d49-228be087c88c · Medium

As a monthly newsletter formed in 1969 by a group of Asian American students at the University of California, Los Angeles, *Gidra* helped raise awareness about social and political issues concerning the Asian American community on campus and at large. The newsletter had an expansive reach for a publication of its kind: around 4,000 copies were published each month. A student writing a history paper, however, hypothesizes that *Gidra*'s influence cannot be measured by the number of newsletters published monthly alone.

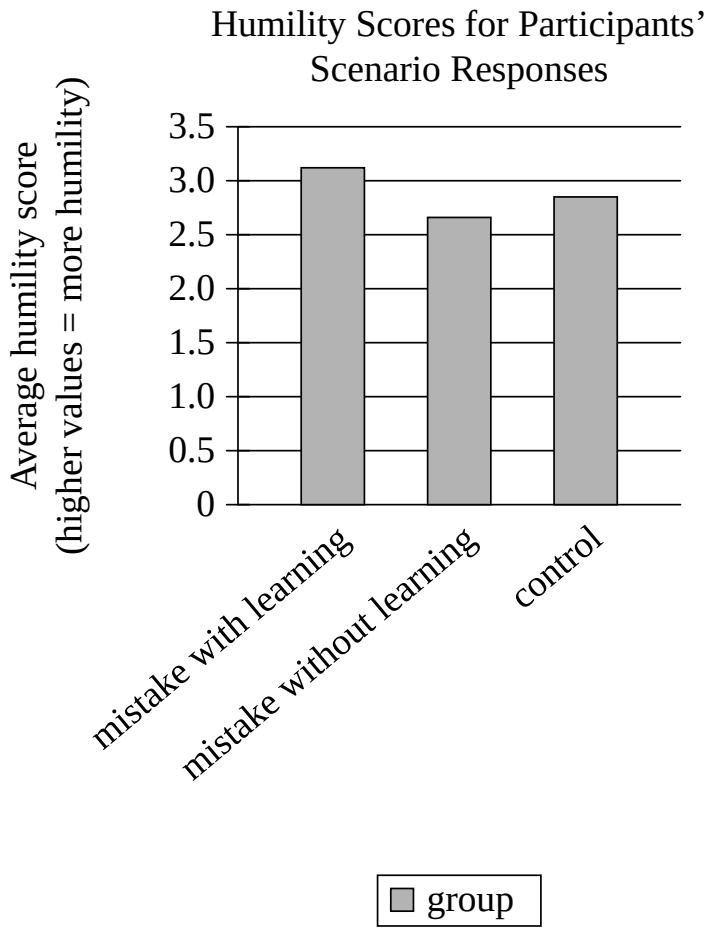
Which finding, if true, would most directly support the student's hypothesis?

A. The students who initially formed *Gidra* each contributed financially to its creation.

B. In addition to covering current events, *Gidra* also featured works of art and literature.

C. *Gidra* was initially based out of the Asian American Studies Center at UCLA.

D. People would often give their copies of *Gidra* to others once they had finished reading an issue.



- The Average humility score data for the 3 bars are as follows:
  - mistake with learning: 3.12
  - mistake without learning: 2.66
  - control: 2.85

Jia Hu and colleagues hypothesized that workplace leaders who reflect on lessons learned from past mistakes are likely to exhibit more humility than leaders who don't engage in such reflection. To test this, the team placed 301 managers in one of three groups. Participants in two experimental groups were asked to reflect on a mistake, one group focusing on a mistake that resulted in learning and the other group focusing on a mistake that didn't result in learning. Participants in a control group were asked to reflect on their daily routine. All the participants then described how they would respond to a workplace scenario. After evaluating the responses for evidence of humility, the researchers concluded that their hypothesis was correct.

Which choice best describes data in the graph that support the researchers' conclusion?

A. None of the three groups' average humility scores exceeded 3.5.

B. The managers in the control group exhibited only slightly less humility on average than the managers in the two experimental groups did.

C. The managers who reflected on a past mistake that resulted in learning exhibited more humility on average than the managers in the other two groups did.

D. All three groups exhibited less humility on average than the researchers expected.

2.58 7d4a4c83-50ae-47ab-9d3d-a67b0cc584c2 · Medium

There Is Confusion is a 1924 novel by Jessie Redmon Fauset. In the novel, the narrator portrays the character Joanna as someone who admires ambition in other people to the exclusion of all other qualities: _____blank

Which quotation from *There Is Confusion* most effectively illustrates the claim?

A. "Joanna was mightily interested in people who had a 'purpose' in life. Otherwise not at all."

B. "Indeed from the very beginning Joanna showed her preference for her father."

C. "Joanna was like her father not only so far as ambition was concerned but also in her willingness to work."

D. "She had a good sense of logic, a strong power of concentration, and a remarkably retentive and visualizing memory."

2.59 bce1433e-0679-4c4f-88e2-a37d1892537d · Medium

External shopping cues are a type of marketing that uses obvious messaging—a display featuring a new product, for example, or a “buy one, get one free” offer—to entice consumers to make spontaneous purchases. In a study, data scientist Sam K. Hui and colleagues found that this effect can also be achieved with a less obvious cue: rearranging a store’s layout. The researchers explain that trying to find items in new locations causes shoppers to move through more of the store, exposing them to more products and increasing the likelihood that they’ll buy an item they hadn’t planned on purchasing.

Which response from a survey given to shoppers who made a purchase at a retail store best supports the researchers’ explanation?

A. “I needed to buy some cleaning supplies, but they weren’t in their regular place. While I was looking for them, I saw this interesting notebook and decided to buy it, too.”

B. “I didn’t buy everything on my shopping list today. I couldn’t find a couple of the items in the store, even though I looked all over for them.”

C. “The store sent me a coupon for a new brand of soup, so I came here to find out what kinds of soup that brand offers. I decided to buy a few cans because I had the coupon.”

D. “This store is larger than one that’s closer to where I live, and it carries more products. I came here to buy some things that the other store doesn’t always have.”

2.60 ef9499bb-6d7f-4c00-a193-e4c998f973b6 · Medium

In addition to her technical skill and daring feats, American stunt pilot Bessie Coleman was also known for dazzling the crowds that came to watch her air shows in the 1920s with her exuberant personality. During her career, she was careful and purposeful about how she crafted her public persona. An aviation researcher has claimed that Coleman intentionally defied social norms of the time by how she chose to present herself to the public.

Which quotation from an article about Coleman would most directly support the aviation researcher's claim?

A. "For her air shows, Coleman frequently used the Curtiss JN-4, or 'Jenny,' which at that time was one of the most well-known types of planes."

B. "While Coleman was beloved by spectators for her charisma, she had a more complicated relationship with her managers and staff, who at times found her behavior too impulsive and demanding."

C. "Coleman once considered leaving her career as a stunt pilot to focus her efforts on giving speeches, which she felt would better support her public image."

D. "Although female pilots were typically expected to wear traditional but impractical attire that included dresses or skirts, photographs of Coleman show her wearing pants and leather jackets."

North American Thrasher Mean Bill Size and Habitat Temperature Range

Species	Mean bill surface area (cm^2cm squared)	Mean maximum temperature of warmest month (°C)	Mean minimum temperature of coldest month (°C)
Brown thrasher	1.86	30.40	−4.29
Bendire's thrasher	1.98	36.57	0.24
Long-billed thrasher	2.24	35.27	8.82
Cozumel thrasher	2.28	33.27	18.21
Ocellated thrasher	3.26	27.56	5.45

It has been hypothesized that since birds can dissipate excess heat through their bills, bill size should increase with habitat temperature. To evaluate this hypothesis for a 2021 study, Charlotte Probst and colleagues gathered data on mean bill surface area of species of North American thrashers (genus: *Toxostoma*) as well as on climate conditions of the birds' native habitats. Based on their data, Probst and colleagues concluded that the hypothesis was not fully supported.

Which choice most effectively uses data from the table to support Probst and colleagues' conclusion?

A. Although the Bendire's thrasher has one of the smallest mean bill surface areas of the birds included in the table, its habitat has one of the lowest mean maximum temperatures in the warmest month and one of the lowest mean minimum temperatures in the coldest month.

B. Although the Cozumel thrasher has the second greatest mean bill surface area of the birds included in the table, its habitat's mean temperature in the warmest month is significantly higher than that of the other birds' habitats.

C. Of the birds included in the table, the brown thrasher has the smallest mean bill surface area and the habitat with the lowest mean minimum temperature in the coldest month, while the

long-billed thrasher has the second-largest mean bill surface area and the habitat with the second-highest mean minimum temperature in the coldest month.

D. Of the birds included in the table, the ocellated thrasher has the largest mean bill surface area and the habitat with the lowest mean maximum temperature in the warmest month, while the Bendire's thrasher has the second smallest mean bill surface area and the habitat with the highest mean maximum temperature in the warmest month.

2.62 c1854738-41cb-44ba-be7e-081b7dfd9993 · Medium

Hedda Gabler is an 1890 play by Henrik Ibsen. As a woman in the Victorian era, Hedda, the play's central character, is unable to freely determine her own future. Instead, she seeks to influence another person's fate, as is evident when she says to another character, _____blank

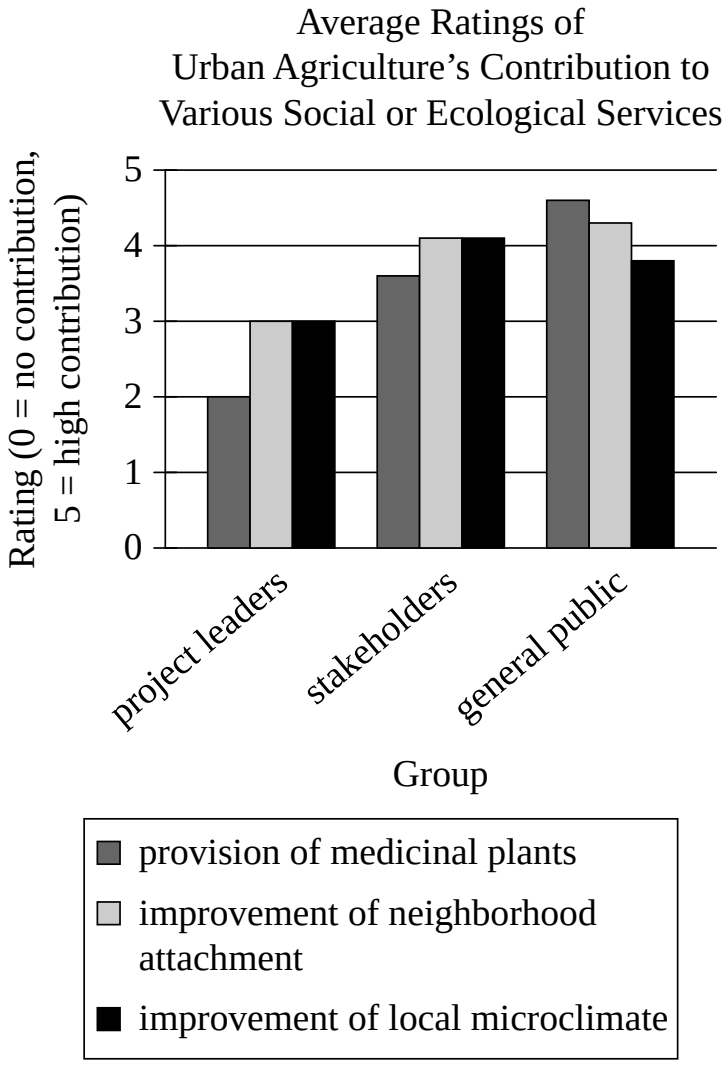
Which quotation from a translation of *Hedda Gabler* most effectively illustrates the claim?

A. "Then what in heaven's name would you have me do with myself?"

B. "I want for once in my life to have power to mould a human destiny."

C. "Then I, poor creature, have no sort of power over you?"

D. "Faithful to your principles, now and for ever! Ah, that is how a man should be!"



- For each data category, the following bars are shown:
  - provision of medicinal plants
  - improvement of neighborhood attachment
  - improvement of local microclimate
- The data for the 3 categories are as follows:
  - project leaders:
    - provision of medicinal plants: 2.0
    - improvement of neighborhood attachment: 3.0
    - improvement of local microclimate: 3.0
  - stakeholders:
    - provision of medicinal plants: 3.6
    - improvement of neighborhood attachment: 4.1

        - improvement of local microclimate: 4.1
    - general public:
        - provision of medicinal plants: 4.6
        - improvement of neighborhood attachment: 4.3
        - improvement of local microclimate: 3.8

Urban agriculture is the practice of growing plant- or animal-based products in urban settings such as community gardens and rooftop farms. Esther Sanyé-Mengual, Kathrin Specht, and their team surveyed three groups of people in Bologna, Italy—leaders of urban agriculture projects, stakeholders in urban agriculture (e.g., food researchers and urban farming associations), and the general public—to compare their views about the extent to which urban agriculture contributes to 25 social or ecological services that the team identified. The survey results show that, on average, project leaders rated urban agriculture as contributing less to _____blank

Which choice most effectively uses data from the graph to complete the assertion?

A. providing medicinal plants than to improving the local microclimate.

B. improving community members' sense of attachment to the neighborhood than to offering educational opportunities.

C. improving the local microclimate than to providing medicinal plants.

D. improving the local microclimate than to providing tourist attractions.

2.64 a7544373-0f0e-48ad-bf64-b636aa0c867e · Hard

Although it's clear that Mars once had liquid water on its surface, astronomers have debated whether the evidence of ancient water reflects a prolonged phase of warm, wet conditions—the so-called wet and warm scenario—or a brief period of melting in an otherwise consistently frozen environment. Researchers Benjamin T. Cardenas and Michael P. Lamb recently added to this debate by using data from NASA and the Mars Orbiter Laser Altimeter to map the topography of what is now a large basin in Mars's northern hemisphere. Cardenas and Lamb concluded that the wet and warm scenario is likely correct.

Which finding about the basin, if true, would most directly support Cardenas and Lamb's conclusion?

A. Its dimensions and shape indicate that it is unlikely to have formed as the result of an asteroid or comet impact.

B. It has features suggesting that it once held an ocean that underwent gradual sea-level changes over an extended time.

C. Its physical characteristics are most consistent with it having formed as a result of a massive but short-lived influx of liquid water.

D. It is surrounded by channels that could have been formed either by running water or by flowing lava.

2.65 a34fe949-4c0c-463d-a566-a28ea8069426 · Hard

Jean-Bernard Caron and colleagues recently discovered a cache of jellyfish fossils in the Burgess Shale, a site in the Canadian Rockies that is rich in fossils from the Cambrian period (over 500 million years ago). Caron and colleagues claim that these are the oldest jellyfish fossils ever discovered. In the past twenty years, two sites in China and the United States have yielded fossils of a similar age that some experts believe are most likely jellyfish due to their shapes and the appearance of projecting tentacles. But Caron and colleagues argue that the apparent tentacles are in fact the comb rows of ctenophores, gelatinous animals that are only distantly related to jellyfish.

Which statement, if true, would most directly weaken the claim by Caron and colleagues about the fossils found in China and the United States?

A. Sites in the Canadian Rockies from later periods than the Cambrian period have yielded fossils that have been conclusively identified as ctenophore fossils.

B. The fossils found in China and the United States are so poorly preserved that though they cannot be conclusively identified as jellyfish, they cannot be conclusively identified as ctenophores either.

C. While ctenophore fossils have been discovered in China and the United States, they have never been discovered in the Burgess Shale.

D. The fossils discovered by Caron and colleagues in the Burgess Shale were better preserved than the fossils discovered by other researchers in China and the United States.

2.66 15eb5d10-e935-4161-bec5-d9798b5f47e3 · Hard

Swahili Speakers in Three African Countries

Country	Approximate number of speakers (in millions)	Estimated % of population
Democratic Republic of the Congo	22	25
Kenya	55	100
Tanzania	61	100

Swahili is estimated to be the first language of up to 15 million people worldwide. It's also an officially recognized language in Tanzania, Kenya, and the Democratic Republic of the Congo, which means these countries use Swahili in government documents and proceedings. But even in countries where almost everyone speaks Swahili, for many it isn't their first language but is instead their second, third, or even fourth language.

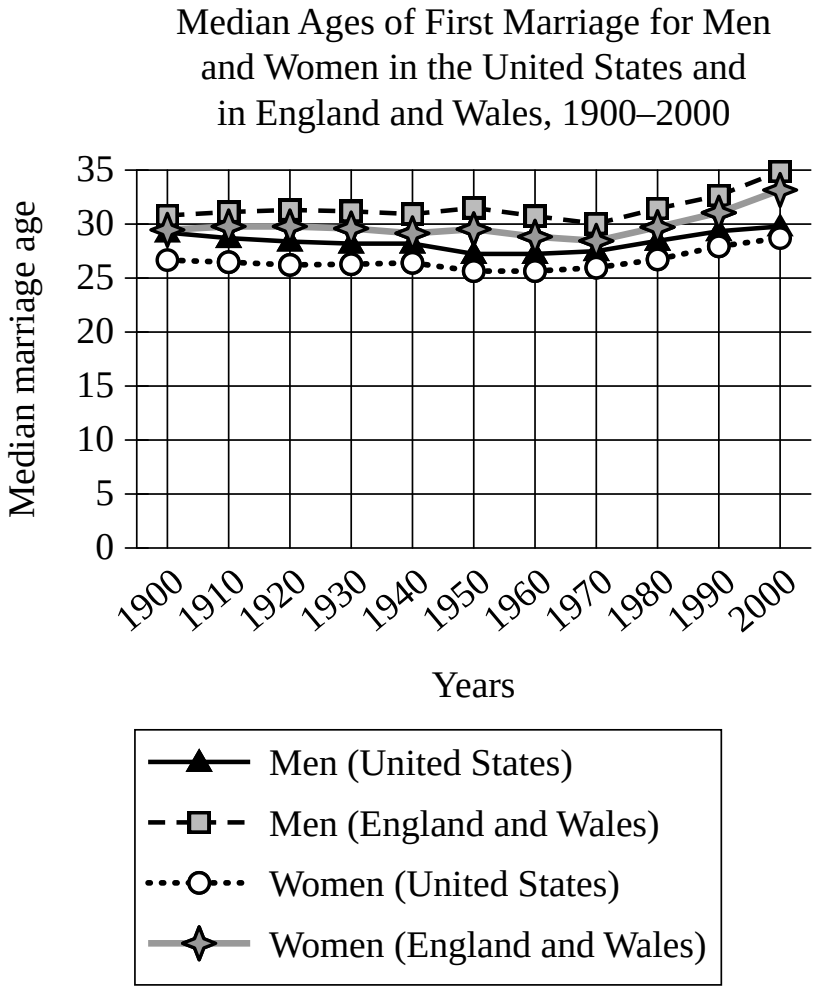
Which choice most effectively uses data from the table to support the underlined claim?

A. Tanzania has approximately 61 million Swahili speakers, which is much more than the estimated total number of people worldwide for whom Swahili is their first language.

B. Tanzania is estimated to have at most 15 million Swahili speakers, while the country's total population is approximately 61 million people.

C. Approximately 100 percent of the people who speak Swahili as their first language live in Kenya, which has a total population of approximately 55 million people.

D. Approximately 100 percent of Kenya's population speaks Swahili, while only about 25 percent of the Democratic Republic of the Congo's population speaks Swahili.



- The following 4 lines are shown:
  - Men (United States)
  - Men (England and Wales)
  - Women (United States)
  - Women (England and Wales)
- The Men (United States) line:
  - Begins at 1900, 25.9
  - Falls gradually to 1910, 25.1
  - Falls gradually to 1920, 24.6
  - Falls gradually to 1930, 24.3
  - Remains level to 1940, 24.3
  - Falls gradually to 1950, 22.8
  - Remains level to 1960, 22.8
  - Rises gradually to 1970, 23.2

    - Rises gradually to 1980, 24.7
    - Rises gradually to 1990, 26.1
    - Rises gradually to 2000, 26.8
- The Men (England and Wales) line:
    - Begins at 1900, 28.4
    - Rises gradually to 1910, 28.9
    - Rises gradually to 1920, 29.2
    - Falls gradually to 1930, 29.0
    - Falls gradually to 1940, 28.6
    - Rises gradually to 1950, 29.5
    - Falls gradually to 1960, 28.3
    - Falls gradually to 1970, 27.2
    - Rises gradually to 1980, 29.4
    - Rises gradually to 1990, 31.3
    - Rises gradually to 2000, 34.8
- The Women (United States) line:
    - Begins at 1900, 21.9
    - Falls gradually to 1910, 21.6
    - Falls gradually to 1920, 21.2
    - Rises gradually to 1930, 21.3
    - Rises gradually to 1940, 21.5
    - Falls gradually to 1950, 20.3
    - Remains level to 1960, 20.3
    - Rises gradually to 1970, 20.8
    - Rises gradually to 1980, 22.0
    - Rises gradually to 1990, 23.9
    - Rises gradually to 2000, 25.1
- The Women (England and Wales) line:
    - Begins at 1900, 26.3
    - Rises gradually to 1910, 26.8
    - Remains level to 1920, 26.8
    - Falls gradually to 1930, 26.5
    - Falls gradually to 1940, 25.8
    - Rises gradually to 1950, 26.4
    - Falls gradually to 1960, 25.3
    - Falls gradually to 1970, 24.7
    - Rises gradually to 1980, 26.7
    - Rises gradually to 1990, 28.8
    - Rises gradually to 2000, 32.1

A sociology student is reading an essay on the median age of first marriage in Western countries throughout the twentieth century. The author of the essay cites factors common to these countries that the author believes caused an increase in the median age of first marriage, such as new technologies that shortened the time needed for domestic chores, making two-person households less necessary and living alone more viable. The student asserts that beyond these factors there must be additional ones specific to particular Western countries that influenced the increase of age at first marriage.

Which choice most effectively uses data from the graph that support the student's assertion?

A. Between 1970 and 2000, the median age of first marriage rose more sharply for men in England and Wales than it did for men in the United States.

B. The median age of first marriage for men in England and Wales was lower in 1970 than in 1950 or 1990.

C. Between 1900 and 2000, the median age of first marriage for women in England and Wales was consistently higher than for women in the United States, as was the case for men.

D. In England and Wales, the median age of first marriage was consistently higher for men than for women between 1900 and 2000, but this was not always the case in the United States.

2.68 f7ae1a33-b06f-4c83-846e-8d697f444fe0 · Hard

Tadpole Body Mass and Toxin Production after Three Weeks in Ponds

Population density	Average tadpole body mass (milligrams)	Average number of distinct bufadienolide toxins per tadpole	Average amount of bufadienolide per tadpole (nanograms)	Average bufadienolide concentration (nanograms per milligram of tadpole body mass)
High	193.87	22.69	5,815.51	374.22
Medium	254.56	21.65	5,525.72	230.10
Low	258.97	22.08	4,664.99	171.43

Ecologist Veronika Bókony and colleagues investigated within-species competition among common toads (*Bufo bufo*), a species that secretes various unpleasant-tasting toxins called bufadienolides in response to threats. The researchers tested *B. bufo* tadpoles' responses to different levels of competition by creating ponds with different tadpole population densities but a fixed amount of food. Based on analysis of the tadpoles after three weeks, the researchers concluded that increased competition drove bufadienolide production at the expense of growth.

Which choice uses data from the table to most effectively support the researchers' conclusion?

A. The difference in average tadpole body mass was small between the low and medium population density conditions and substantially larger between the low and high population density conditions.

B. Tadpoles in the low and medium population density conditions had substantially lower average bufadienolide concentrations but had greater average body masses than those in the high population density condition.

C. Tadpoles in the high population density condition displayed a relatively modest increase in the average amount of bufadienolide but roughly double the average bufadienolide concentration compared to those in the low population density condition.

D. Tadpoles produced approximately the same number of different bufadienolide toxins per individual across the population density conditions, but average tadpole body mass decreased as population density increased.

2.69 ccfafc3d-5b74-4194-a041-8bc4d7116868 · Hard

Born in Chile in 1917, artist and ethnomusicologist Violeta Parra was a pioneer in the *nueva canción chilena* (Chilean New Song) movement that emerged in the late 1950s and then spread throughout Latin America, Portugal, and Spain as *nueva canción*. Parra traveled all over Chile compiling extensive records of authentic folk music as well as recipes, proverbs, and other facets of cultural history. These records formed the foundation for the early movement's revival of traditional Chilean folk forms in new songs that represented modern realities of the working class and strongly advocated for social change. As the movement spread beyond Chile, the breadth of musical traditions incorporated into its foundation also expanded.

Which detail about songs associated with *nueva canción*, if true, would best illustrate the underlined claim?

A. Many feature political commentary addressing contemporary issues that stemmed from shared experiences of European colonization in Latin American countries.

B. Many demonstrate the stylistic influence of *corrido*, a genre of narrative songs from Mexico that had come to be characterized by political themes in the early 1800s.

C. Many were written with parts meant to be played on the *quena*, a traditional flute used across Andean countries, including Chile.

D. Many were produced by Argentinian artists in the late 1950s, with others by artists in additional Latin American countries first emerging soon after.

2.70 79e4ee43-9828-46ac-afcb-d3678e7e0d1c · Hard

An archaeological team led by Piotr Bieliński and Sultan al-Bakri found remnants of a 4,000-year-old Bronze Age board game at a site in Oman. Little is left of the game except a stone board, which is carved with a grid and has places to hold game pieces. Some scholars claim that the game was largely played by traders.

Which finding, if true, would most directly support the scholars' claim?

A. Other examples of the game dating to the same period have been found in the remains of several homes in the region, including in one home that may have belonged to a trader.

B. Similar games have been found in other sites dating to the same period that were connected to the site in Oman via trade routes.

C. The other known examples of the game dating to the same period have been found along routes that seem to have been used primarily by traders at the time.

D. Remnants of other goods have been found at the site in Oman that probably also reached the location through trade.

2.71 f445072c-007e-485c-a469-b2b4bee641b3 · Hard

Mosasaurs were large marine reptiles that lived in the Late Cretaceous period, approximately 100 million to 66 million years ago. Celina Suarez, Alberto Pérez-Huerta, and T. Lynn Harrell Jr. examined oxygen-18 isotopes in mosasaur tooth enamel in order to calculate likely mosasaur body temperatures and determined that mosasaurs were endothermic—that is, they used internal metabolic processes to maintain a stable body temperature in a variety of ambient temperatures. Suarez, Pérez-Huerta, and Harrell claim that endothermy would have enabled mosasaurs to include relatively cold polar waters in their range.

Which finding, if true, would most directly support Suarez, Pérez-Huerta, and Harrell's claim?

A. Mosasaurs' likely body temperatures are easier to determine from tooth enamel oxygen-18 isotope data than the body temperatures of nonendothermic Late Cretaceous marine reptiles are.

B. Fossils of both mosasaurs and nonendothermic marine reptiles have been found in roughly equal numbers in regions known to be near the poles during the Late Cretaceous, though in lower concentrations than elsewhere.

C. Several mosasaur fossils have been found in regions known to be near the poles during the Late Cretaceous, while relatively few fossils of nonendothermic marine reptiles have been found in those locations.

D. During the Late Cretaceous, seawater temperatures were likely higher throughout mosasaurs' range, including near the poles, than seawater temperatures at those same latitudes are today.

2.72 8ce518f3-dd51-4bc0-8eb6-72b9a8219145 · Hard

In a research paper, a student criticizes some historians of modern African politics, claiming that they have evaluated Patrice Lumumba, the first prime minister of what is now the Democratic Republic of the Congo, primarily as a symbol rather than in terms of his actions.

Which quotation from a work by a historian would best illustrate the student's claim?

A. "Lumumba is a difficult figure to evaluate due to the starkly conflicting opinions he inspired during his life and continues to inspire today."

B. "The available information makes it clear that Lumumba's political beliefs and values were largely consistent throughout his career."

C. "Lumumba's practical accomplishments can be passed over quickly; it is mainly as the personification of Congolese independence that he warrants scholarly attention."

D. "Many questions remain about Lumumba's ultimate vision for an independent Congo; without new evidence coming to light, these questions are likely to remain unanswered."

2.73 857c3b1f-07bb-491a-84c5-c45a5f28b850 · Hard

Employment by Sector in France and the United States, 1800–2012 (% of total employment)

Year	Agriculture in France	Manufacturing in France	Services in France	Agriculture in US	Manufacturing in US	Services in US
1800	64	22	14	68	18	13
1900	43	29	28	41	28	31
1950	32	33	35	14	33	53
2012	3	21	76	2	18	80

Rows in table may not add up to 100 due to rounding.

Over the past two hundred years, the percentage of the population employed in the agricultural sector has declined in both France and the United States, while employment in the service sector (which includes jobs in retail, consulting, real estate, etc.) has risen. However, this transition happened at very different rates in the two countries. This can be seen most clearly by comparing the employment by sector in both countries in _____blank

Which choice most effectively uses data from the table to complete the statement?

A. 1900 with the employment by sector in 1950.

B. 1800 with the employment by sector in 2012.

C. 1900 with the employment by sector in 2012.

D. 1800 with the employment by sector in 1900.

2.74 047e7067-f570-44e5-8e44-a92ea3f8b3b9 · Hard

Neurobiologists Laura Cuaya, Raúl Hernández-Pérez, and colleagues investigated the language detection abilities of eighteen dogs. The researchers monitored the brain activity of Joey (an Australian shepherd), Mini (a mixed breed), and other dogs while the animals listened to three recordings: one of *The Little Prince* being read in Spanish, the second in Hungarian, and a third made up of short, randomly selected fragments of the first two, scrambled so that they didn't resemble human speech. Each dog was familiar with either Spanish or Hungarian, but not both. The team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech.

Which finding from the study, if true, would most directly support the team's conclusion?

A. Long-headed dogs accustomed to hearing Spanish tended to show more brain activity when hearing Spanish than long-headed dogs accustomed to hearing Hungarian showed when hearing Hungarian.

B. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing either Spanish or Hungarian than when hearing the scrambled recording.

C. The pattern of brain activity that long-headed dogs showed when hearing the scrambled recording was different from the pattern of brain activity that short-headed dogs showed when hearing the language they were accustomed to.

D. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing the language they were accustomed to than when hearing the other language.

2.75 95a5360b-3135-4780-921f-c511f0ae2578 · Hard

In 1534 CE, King Henry VIII of England split with the Catholic Church and declared himself head of the Church of England, in part because Pope Clement VII refused to annul his marriage to Catherine of Aragon. Two years later, Henry VIII introduced a policy titled the Dissolution of the Monasteries that by 1540 had resulted in the closure of all Catholic monasteries in England and the confiscation of their estates. Some historians assert that the enactment of the policy was primarily motivated by perceived financial opportunities.

Which quotation from a scholarly article best supports the assertion of the historians mentioned in the text?

A. “At the time of the Dissolution of the Monasteries, about 2 percent of the adult male population of England were monks; by 1690, the proportion of the adult male population who were monks was less than 1 percent.”

B. “A contemporary description of the Dissolution of the Monasteries, Michael Sherbrook’s *Falle of the Religious Howses*, recounts witness testimony that monks were allowed to keep the contents of their cells and that the monastery timber was purchased by local yeomen.”

C. “In 1535, the year before enacting the Dissolution of the Monasteries, Henry commissioned a survey of the value of church holdings in England—the work, performed by sheriffs, bishops, and magistrates, began that January and was swiftly completed by the summer.”

D. “The October 1536 revolt known as the Pilgrimage of Grace had several economic motives: high food prices due to a poor harvest the prior year; the Dissolution of the Monasteries, which closed reliable sources of food and shelter for many; and rents and taxes throughout Northern England that were not merely high but predatory.”

2.76 17089083-33c6-4d4c-9f5d-eb517d0a9341 · Hard

Gorgets, or necklaces with large pendants, have been part of the ceremonial attire of tribes from the US Southeast for centuries. One of the oldest examples, the Fairfield Gorget, was found in Fairfield, Missouri, in 1958. Its overall design resembles that of other art from the region during the Mississippian period (900–1600 CE). Yet the image on the gorget is of a jaguar—a species whose range doesn't extend to Missouri. Jaguar images are common in ancient Mexican art, and Mexico lies squarely in the species' range. Therefore, some scholars argue that long-distance trade in Mexican art objects brought the imagery to Missouri, where a local artist could have adopted it and incorporated it into the Fairfield Gorget.

Which finding, if true, would most strongly support the underlined explanation?

A. The Fairfield Gorget is dated not to the Mississippian period but instead to the earlier Woodland period, which ended around 900 CE.

B. The range of the jaguar is shown to have expanded dramatically after the Mississippian period came to a close around 1600 CE.

C. An ancient Mexican art object is found at a site that dates to the Mississippian period and is close to where the Fairfield Gorget was found.

D. Certain works by present-day artists from Southeastern tribes reflect outside cultural influences, including contemporary Mexican art.

2.77 c3552a8a-a216-4405-9af2-f9ebd746c320 · Hard

Richard II is a play from the 1590s by William Shakespeare. Although King Richard has been vanquished by his cousin Henry Bolingbroke, he intimates that he is not entirely ready to show subservience to his cousin, saying, _____blank

Which quotation from *Richard II* most effectively illustrates the claim?

A. “I have no name, no title,— / No, not that name was given me at the font,— / But ’tis usurp’d:— Alack the heavy day, / That I have worn so many winters out, / And know not now what name to call myself!”

B. “Alack, why am I sent for to a king, / Before I have shook off the regal thoughts / Wherewith I reign’d? I hardly yet have learn’d / To insinuate, flatter, bow, and bend my knee.”

C. “Still my griefs are mine. / You may my glories and my state depose, / But not my griefs; still am I king of those.”

D. “Am I both priest and clerk? Well then, amen. / God save the King! although I be not he; / And yet, amen, if heaven do think him me.”

2.78 35ba1503-3ca4-4444-aa35-a1bfdee283cf · Hard

The novelist Toni Morrison was the first Black woman to work as an editor at the publishing company Random House, from 1967 to 1983. A scholar asserts that one of Morrison’s likely aims during her time as an editor was to strengthen the presence of Black writers on the list of Random House’s published authors.

Which finding, if true, would most strongly support the scholar’s claim?

A. The percentage of authors published by Random House who were Black rose in the early 1970s and stabilized throughout the decade.

B. Black authors who were interviewed in the 1980s and 1990s were highly likely to cite Toni Morrison’s novels as a principal influence on their work.

C. The novels written by Toni Morrison that were published after 1983 sold significantly more copies and received wider critical acclaim than the novels she wrote that were published before 1983.

D. Works that were edited by Toni Morrison during her time at Random House displayed stylistic characteristics that distinguished them from works that were not edited by Morrison.

2.79 68948483-1f64-4760-a657-ec2406b34c19 · Hard

Political scientists who favor the traditional view of voter behavior claim that voting in an election does not change a voter's attitude toward the candidates in that election. Focusing on each US presidential election from 1976 to 1996, Ebonya Washington and Sendhil Mullainathan tested this claim by distinguishing between subjects who had just become old enough to vote (around half of whom actually voted) and otherwise similar subjects who were slightly too young to vote (and thus none of whom voted). Washington and Mullainathan compared the attitudes of the groups of subjects toward the winning candidate two years after each election.

Which finding from Washington and Mullainathan's study, if true, would most directly weaken the claim made by people who favor the traditional view of voter behavior?

A. Subjects' attitudes toward the winning candidate two years after a given election were strongly predicted by subjects' general political orientation, regardless of whether subjects were old enough to vote at the time of the election.

B. Subjects who were not old enough to vote in a given election held significantly more positive attitudes towards the winning candidate two years later than they held at the time of the election.

C. Subjects who voted in a given election held significantly more polarized attitudes toward the winning candidate two years later than did subjects who were not old enough to vote in that election.

D. Two years after a given election, subjects who voted and subjects who were not old enough to vote were significantly more likely to express negative attitudes than positive attitudes toward the winning candidate in that election.

2.80 2ec9a135-3760-47a4-9f8c-a21f7e5dee3e · Hard

Credited Film Output of James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr

Individual	Years active	Number of films known and commonly credited
James Young Deer	1909–1924	33 (actor), 35 (director), 10 (writer)
Dark Cloud	1910–1920	35 (actor), 1 (writer)
Edwin Carewe	1912–1934	47 (actor), 58 (director), 20 (producer), 4 (writer)
Lillian St. Cyr (Red Wing)	1908–1921	66 (actor)

Some researchers studying Indigenous actors and filmmakers in the United States have turned their attention to the early days of cinema, particularly the 1910s and 1920s, when people like James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr (known professionally as Red Wing) were involved in one way or another with numerous films. In fact, so many films and associated records for this era have been lost that counts of those four figures' output should be taken as bare minimums rather than totals; it's entirely possible, for example, that _____blank

Which choice most effectively uses data from the table to complete the example?

A. Dark Cloud acted in significantly fewer films than did Lillian St. Cyr, who is credited with 66 performances.

B. Edwin Carewe's 47 credited acting roles includes only films made after 1934.

C. Lillian St. Cyr acted in far more than 66 films and Edwin Carewe directed more than 58.

D. James Young Deer actually directed 33 films and acted in only 10.

2.81 eec70d3b-a52b-4324-9e00-8799afda175a · Hard

Psychologists Dacher Keltner and Jonathan Haidt have argued that experiencing awe—a sensation of reverence and wonder typically brought on by perceiving something grand or powerful—can enable us to feel more connected to others and thereby inspire us to act more altruistically. Keltner, along with Paul K. Piff, Pia Dietze, and colleagues, claims to have found evidence for this effect in a recent study where participants were asked to either gaze up at exceptionally tall trees in a nearby grove (reported to be a universally awe-inspiring experience) or stare at the exterior of a nearby, nondescript building. After one minute, an experimenter deliberately spilled a box of pens nearby.

Which finding from the researchers' study, if true, would most strongly support their claim?

A. Participants who had been looking at the trees helped the experimenter pick up significantly more pens than did participants who had been looking at the building.

B. Participants who helped the experimenter pick up the pens used a greater number of positive words to describe the trees and the building in a postexperiment survey than did participants who did not help the experimenter.

C. Participants who did not help the experimenter pick up the pens were significantly more likely to report having experienced a feeling of awe, regardless of whether they looked at the building or the trees.

D. Participants who had been looking at the building were significantly more likely to notice that the experimenter had dropped the pens than were participants who had been looking at the trees.

2.82 15e61784-d141-4360-95ad-6de4a72b0a0d · Hard

Number and Origin of Clamshell Tools Found at Different Levels Below the Surface in Neanderthal Cave

Depth of tools found below surface in cave (meters)	Clamshells that Neanderthals collected from the beach	Clamshells that Neanderthals harvested from the seafloor
3–4	99	33
6–7	1	0
4–5	2	0
2–3	7	0
5–6	18	7

Studying tools unearthed at a cave site on the western coast of Italy, archaeologist Paola Villa and colleagues have determined that prehistoric Neanderthal groups fashioned them from shells of clams that they harvested from the seafloor while wading or diving or that washed up on the beach. Clamshells become thin and eroded as they wash up on the beach, while those on the seafloor are smooth and sturdy, so the research team suspects that Neanderthals prized the tools made with seafloor shells. However, the team also concluded that those tools were likely more challenging to obtain, noting that _____blank

Which choice most effectively uses data from the table to support the research team's conclusion?

A. at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that shells were easier to collect from the beach than to harvest from the seafloor.

B. the highest number of tools were at a depth of 3–4 meters below the surface, which suggests that the Neanderthal population at the site was highest during the related period of time.

C. at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that Neanderthals preferred to use clamshells from the beach because of their durability.

D. the higher number of tools at depths of 5–6 meters below the surface in the cave than at depths of 4–5 meters below the surface suggests that the size of clam populations changed

over time.

2.83 0ce08800-10d9-4d16-a1c0-8564259c6e1f · Hard

Initially observed in 2017, the interstellar object ‘Oumuamua is the first object of its kind to be seen in our solar system. Researchers have been puzzled because its acceleration cannot be entirely explained by the gravitational pull of nearby bodies: there must be a nongravitational influence on its velocity and trajectory. Some previously suggested explanations for this nongravitational acceleration involve mechanisms that are unlikely or unrealistic, such as geometric effects from ‘Oumuamua being potentially composed of several spatially separated bodies. Now, Jennifer Bergner and colleagues propose that the nongravitational acceleration is due to the gaseous expulsion of entrapped hydrogen from ‘Oumuamua’s water-rich icy body.

Which statement, if true, would most strongly support the claim made by Bergner and colleagues about the cause of ‘Oumuamua’s acceleration?

A. Existing proposed models of outgassing from ‘Oumuamua include the direct conversion of nitrogen or carbon monoxide from a solid to a gaseous state without becoming liquid, but these models have theoretical or observational inconsistencies.

B. ‘Oumuamua’s trajectory is inconsistent with a nongravitational acceleration that would be caused by the release of hydrogen gas resulting from the processing of water ice (H_2O), but the interstellar object’s observable properties can be explained if it has a significant component of molecular hydrogen ice (H_2).

C. Since nongravitational accelerations of interstellar objects are several orders of magnitude weaker than gravitational accelerations, deviation from behavior that could be fully attributed to gravitational pull has been detected on a limited number of objects similar to ‘Oumuamua.

D. Exposure to interstellar cosmic radiation can result in the formation of embedded pockets of hydrogen gas in water ice; moreover, when traveling through the solar system, ‘Oumuamua experienced warming sufficient to alter its icy structure and allow for outgassing.

2.84 d8fe9867-e89f-45bd-b49f-7fcfe7068162 · Hard

In subtropical Asia, *Apis dorsata* (giant honeybee) plays an essential role in pollinating a wide variety of crops and wild plants. To study how different agricultural land covers affect the species, Rika Raffiudin and colleagues monitored the foraging activity of the bees as well as the pollen content of the honey from *A. dorsata* colonies at two sites in Indonesia: Kampar, characterized by its surrounding monoculture farms (growing a single crop), and Kerinci, a forest-agriculture site where multiple crops, including hot peppers and coffee, are grown nearby. The researchers concluded that a lack of crop variety may reduce total pollen collection by *A. dorsata*.

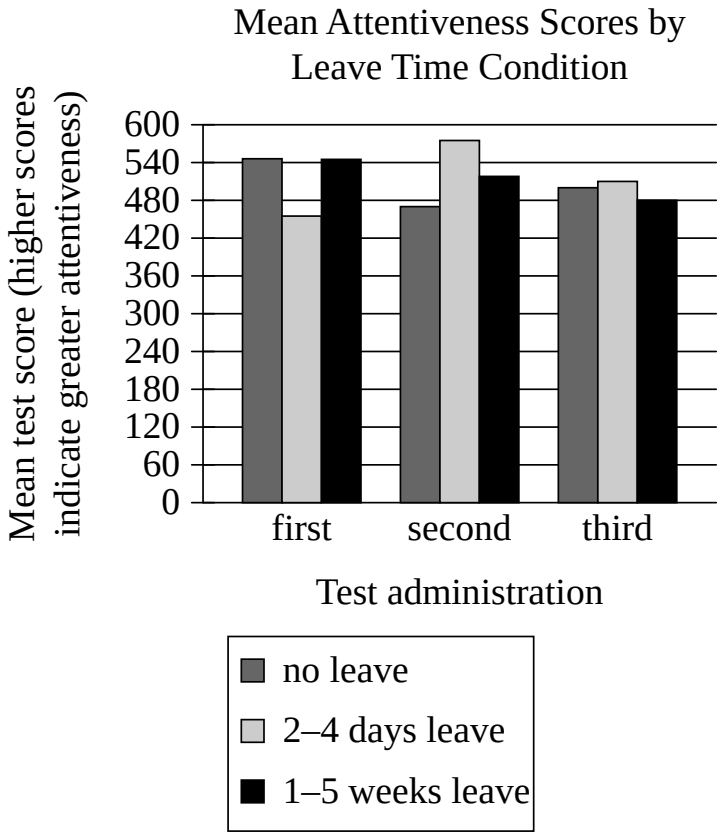
Which finding, if true, would most directly support the researchers' conclusion?

A. Pollen in honey samples from Kampar bee colonies was predominantly sourced from a single plant species, whereas pollen in honey samples from Kerinci bee colonies was sourced from multiple different plant species.

B. In one Kerinci bee colony, a greater proportion of bees returned to their nests with pollen than returned without pollen, whereas the inverse was observed in a second Kerinci bee colony.

C. Significantly fewer bees were observed engaging in foraging activities with the crops surrounding Kerinci than with the crops surrounding Kampar.

D. Honey samples from Kerinci bee colonies contained significantly higher concentrations of pollen than honey samples from Kampar bee colonies did.



- For each data category, the following bars are shown:
  - no leave
  - 2–4 days leave
  - 1–5 weeks leave
- The data for the 3 categories are as follows:
  - First test administration:
    - no leave: 546
    - 2–4 days leave: 455
    - 1–5 weeks leave: 545
  - Second test administration:
    - no leave: 470
    - 2–4 days leave: 575
    - 1–5 weeks leave: 518
  - Third test administration:
    - no leave: 500
    - 2–4 days leave: 510
    - 1–5 weeks leave: 480

To investigate potential cognitive benefits of taking leave from work, psychologist Jan Packer and colleagues conducted a six-month study of Australian university staff members who took no leave from work during the study, took 2–4 days of leave, or took 1–5 weeks of leave. Tests of attentiveness were administered to participants three times during the study: at random for the no-leave staff, and for the rest, one week before their leave, one week following their return to work, and one week after the second test administration. After analyzing the results, the researchers concluded that longer leave times might not confer a greater cognitive benefit than shorter leave times do.

Which choice best describes data from the graph that support the researchers' conclusion?

A. In the second test administration, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave, but in the third test administration, those who took no leave had higher average scores than those who took 1–5 weeks of leave.

B. In the first test administration, participants who took 2–4 days of leave had lower average attentiveness scores than did those who took 1–5 weeks of leave and those who took no leave.

C. In both the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did participants who took 1–5 weeks of leave.

D. In the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave.

2.86 bf4990d3-7a3a-49ff-b7dc-8ba4cf811d16 · Hard

Art collectives, like the United States- and Vietnam-based collective The Propeller Group or Cuba's Los Carpinteros, are groups of artists who agree to work together: perhaps for stylistic reasons, or to advance certain shared political ideals, or to help mitigate the costs of supplies and studio space. Regardless of the reasons, art collectives usually involve some collaboration among the artists. Based on a recent series of interviews with various art collectives, an arts journalist claims that this can be difficult for artists who are often used to having sole control over their work.

Which quotation from the interviews best illustrates the journalist's claim?

A. "The first collective I joined included many amazingly talented artists, and we enjoyed each other's company, but because we had a hard time sharing credit and responsibility for our work, the collective didn't last."

B. "We work together, but that doesn't mean that individual projects are equally the work of all of us. Many of our projects are primarily the responsibility of whoever originally proposed the work to the group."

C. "Having worked as a member of a collective for several years, it's sometimes hard to recall what it was like to work alone without the collective's support. But that support encourages my individual expression rather than limits it."

D. "Sometimes an artist from outside the collective will choose to collaborate with us on a project, but all of those projects fit within the larger themes of the work the collective does on its own."

2.87 30d872e5-cff4-46a3-89ab-715e2bf8bdbe · Hard

Having demonstrated that the power conversion efficiency of new perovskite photovoltaic cells is nearly indistinguishable from that of cells recycled multiple times, Xun Xiao and colleagues evaluated the levelized cost of electricity (LCOE)—a measure of cost per produced electricity unit that incorporates materials acquisition costs as well as operating costs—for power plants utilizing each type of cell. With a modeled 15-year cell lifetime, the LCOE of plants using new cells was 4.99 cents per kilowatt-hour, whereas that of plants using cells recycled three times was 4.05 cents per kilowatt-hour. Moreover, the LCOE differential grew from 18.8% to 31.3% when the modeled cell lifetime was reduced by two-thirds.

Based on the text, which choice, if true, most coherently and directly accounts for the observation presented in the underlined sentence?

A. Cells can be recycled multiple times without a significant decrease in power conversion efficiency, and thus recycling cells more frequently does not change the LCOE for plants using recycled cells.

B. Although potentially less expensive than producing new cells, recycling cells does incur costs, and thus reducing cell lifetimes entails greater cumulative costs.

C. Securing new cell materials is more expensive than recycling cell materials, and thus more frequent cell replacement entails greater disparity in cumulative costs.

D. As the modeled cell lifetime decreases, plant operating costs increase, and thus the LCOE for plants using new cells diverges from that of plants using recycled cells.

2.88 44750441-b8b2-4b69-989c-dd7200101899 · Hard

The presence of other individuals of the same species has been observed to mitigate stress in highly social mammals. To investigate whether this phenomenon, known as social buffering, also occurs in reptiles, researchers led by Chelsea E. Martin monitored stress responses in wild southern Pacific rattlesnakes (*Crotalus helleri*) in three experimental treatments: when alone, with a rope, and with a companion *C. helleri*. The researchers compared the percent change between baseline and peak heart rate in response to a (harmless) disturbance, with higher values indicating higher stress levels.

Which finding, if true, would most directly support the idea that social buffering occurs among *C. helleri*?

A. The average percent change in heart rate was lower among *C. helleri* with a companion than among solitary *C. helleri* and *C. helleri* with a rope.

B. Average peak heart rates were highest among solitary *C. helleri*, but no differences were observed in average peak heart rates between *C. helleri* with a companion and *C. helleri* with a rope.

C. *C. helleri* with a companion displayed a lower average baseline heart rate and lower average peak heart rate than did solitary *C. helleri* or *C. helleri* with a rope.

D. Solitary *C. helleri* had higher average baseline heart rates than did *C. helleri* in the other treatments, but the average percent change in heart rate was smaller among solitary *C. helleri* than among *C. helleri* with a companion.

2.89 9d63dd85-3fe2-4008-b144-de2a399f793e · Hard

A student performs an experiment testing her hypothesis that a slightly acidic soil environment is more beneficial for the growth of the plant *Brassica rapa parachinensis* (a vegetable commonly known as choy sum) than a neutral soil environment. She plants sixteen seeds of choy sum in a mixture of equal amounts of coffee grounds (which are highly acidic) and potting soil and another sixteen seeds in potting soil without coffee grounds as the control for the experiment. The two groups of seeds were exposed to the same growing conditions and monitored for three weeks.

Which finding, if true, would most directly weaken the student's hypothesis?

A. The choy sum planted in the soil without coffee grounds were significantly taller at the end of the experiment than the choy sum planted in the mixture of soil and coffee grounds.

B. The choy sum grown in the soil without coffee grounds weighed significantly less at the end of the experiment than the choy sum grown in the mixture of soil and coffee grounds.

C. The choy sum seeds planted in the soil without coffee grounds sprouted significantly later in the experiment than did the seeds planted in the mixture of soil and coffee grounds.

D. Significantly fewer of the choy sum seeds planted in the soil without coffee grounds sprouted plants than did the seeds planted in the mixture of soil and coffee grounds.

2.90 35e63faf-cd1e-42e6-b544-7bf3b48941eb · Hard

In the 1980s, many musicians and journalists in the English-speaking world began to draw attention to music from around the globe—such as mbaqanga from South Africa and quan họ from Vietnam—that can't be easily categorized according to British or North American popular music genres, typically referring to such music as "world music." While some scholars have welcomed this development for bringing diverse musical forms to prominence in countries where they'd previously been overlooked, musicologist Su Zheng claims that the concept of world music homogenizes highly distinct traditions by reducing them all to a single category.

Which finding about mbaqanga and quan họ, if true, would most directly support Zheng's claim?

A. Mbaqanga and quan họ developed independently of each other and have little in common musically.

B. Mbaqanga is significantly more popular in the English-speaking world than quan họ is.

C. Mbaqanga and quan họ are now performed by a diverse array of musicians with no direct connections to South Africa or Vietnam.

D. Mbaqanga and quan họ are highly distinct from British and North American popular music genres but similar to each other.

2.91 6537fc25-1318-49e9-9e1e-dcc07604c519 · Hard

Simulated Change in Annual Aquifer Input and Irrigation Output if Precipitation Concentration Increases as Climate Models Predict

Baseline concentration of annual precipitation	% change in water entering aquifers	% change in surface water used for irrigation	% change in groundwater used for irrigation
Precipitation is currently somewhat concentrated	4.9	0.4	0.9
Precipitation is currently evenly distributed	11.0	9.0	7.9

Some climate models for the western United States predict that while total annual precipitation may remain unchanged from the present level, precipitation will become concentrated into fewer but more intense rain and snow events. University of Texas climate scientist Geeta Persad and her colleagues simulated how the amount of water entering aquifers and the amount being used for irrigation purposes would change if this were to occur. Persad and her colleagues concluded that concentration of precipitation into fewer events would result in a higher number of dry days, triggering more irrigation, but that this change in irrigation output is highly sensitive to the baseline concentration of precipitation that currently exists in an area.

Which choice best describes data from the table that support Persad and her colleagues' conclusion?

A. If baseline precipitation is somewhat concentrated, the amount of water being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater, whereas the amount of water entering aquifers will increase 11.0% if baseline precipitation is evenly distributed.

B. If baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas it will increase 9.0% for surface water and 7.9% for groundwater if baseline precipitation is evenly distributed.

C. If baseline precipitation is somewhat concentrated, the amount of water entering aquifers will increase 4.9%, while the amount being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater.

D. If baseline precipitation is somewhat concentrated, water use for irrigation will decline by a small amount, whereas it will increase 11.0% for surface water and 9.0% for groundwater if baseline precipitation is evenly distributed.

2.92 e46bbf40-c607-49b6-b493-1fb84005ae31 · Hard

Average Number and Duration of Torpor Bouts and Arousal Episodes for Alaska Marmots and Arctic Ground Squirrels, 2008–2011

Feature	Alaska marmots	Arctic ground squirrels
torpor bouts	12	10.5
duration per bout	13.81 days	16.77 days
arousal episodes	11	9.5
duration per episode	21.2 hours	14.2 hours

When hibernating, Alaska marmots and Arctic ground squirrels enter a state called torpor, which minimizes the energy their bodies need to function. Often a hibernating animal will temporarily come out of torpor (called an arousal episode) and its metabolic rate will rise, burning more of the precious energy the animal needs to survive the winter. Alaska marmots hibernate in groups and therefore burn less energy keeping warm during these episodes than they would if they were alone. A researcher hypothesized that because Arctic ground squirrels hibernate alone, they would likely exhibit longer bouts of torpor and shorter arousal episodes than Alaska marmots.

Which choice best describes data from the table that support the researcher's hypothesis?

A. The Alaska marmots' arousal episodes lasted for days, while the Arctic ground squirrels' arousal episodes lasted less than a day.

B. The Alaska marmots and the Arctic ground squirrels both maintained torpor for several consecutive days per bout, on average.

C. The Alaska marmots had shorter torpor bouts and longer arousal episodes than the Arctic ground squirrels did.

D. The Alaska marmots had more torpor bouts than arousal episodes, but their arousal episodes were much shorter than their torpor bouts.

2.93 c4d80f10-e0f8-4466-bc5a-e45343eceed5 · Hard

Although many transposons, DNA sequences that move within an organism's genome through shuffling or duplication, have become corrupted and inactive over time, those from the long interspersed nuclear elements (LINE) family appear to remain active in the genomes of some species. In humans, they are functionally important within the hippocampus, a brain structure that supports complex cognitive processes. When the results of molecular analysis of two species of octopus—an animal known for its intelligence—were announced in 2022, the confirmation of a LINE transposon in *Octopus vulgaris* and *Octopus bimaculoides* genomes prompted researchers to hypothesize that that transposon family is tied to a species' capacity for advanced cognition.

Which finding, if true, would most directly support the researchers' hypothesis?

A. The LINE transposon in *O. vulgaris* and *O. bimaculoides* genomes is active in an octopus brain structure that functions similarly to the human hippocampus.

B. The human genome contains multiple transposons from the LINE family that are all primarily active in the hippocampus.

C. A consistent number of copies of LINE transposons is present across the genomes of most octopus species, with few known corruptions.

D. *O. vulgaris* and *O. bimaculoides* have smaller brains than humans do relative to body size, but their genomes contain sequences from a wider variety of transposon families.